Quantum Fault Tolerance Meets Toom's Contours

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1 Notations

We follow the standard notations commonly used in the literature. For an integer n, we denote by $[n]$ the set of integers $\{1, 2, \ldots, n\}$. To present a countable set, which might or might not be finite, where the number of elements does not matter, we use curly brackets with a subscript denoting the first index of the set. For example, $\{u_i\}_{i=1}$ represents the set $u_1, u_2, \ldots$; For finite sets, we might add a superscript to present the size of the set. We use a colon inside curly brackets to specify a set of elements satisfying a condition, i.e., for sets A, B, the notation $\{a : a \in A, a \in B\}$ equals $A \cap B$. Curly brackets without subscripts or conditions refer to singletons. For example, for a given integer i, the set $\{u_i\}$ is the set that contains only the element u_i. For an integer i, we denote by 1_i the vector consisting of one at the ith coordinate and zero elsewhere.

When not otherwise mentioned, we consider quantum states over qubits, each lying in a two-dimensional Hilbert space $\mathcal{H}_2$, and denote by $\mathcal{H}_{2^n}$ the Hilbert space of an n-qubit system. Pure quantum states and their dual presentations are denoted by $|\psi\rangle, \langle\psi|$, and in the matrix representation, we use $|\psi\rangle\langle\psi|$ to denote its density matrix. We will use ρ to denote a mixed state, whose density matrix is a trace one PSD matrix.

Below, a semi proof that in CSS code partial correction $Z^e \to Z^{\hat{e}}$ doesn't add bit flip errors. It might adds a global phase. I should enter it in a section about CSS coding.

$$\begin{aligned}
& X^f Z^e \left|w + C_z^\perp\right\rangle \\
=& X^f Z^e X^w \left|C_z^\perp\right\rangle \\
\text{apply } H^{\otimes n} \mapsto& Z^f X^e Z^w |C_z\rangle \\
=& (-1)^{e\cdot w} Z^f Z^w X^e |C_z\rangle \\
\text{partial correction } X^{e+\hat{e}} \mapsto& (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} Z^f Z^w X^{\hat{e}} |C_z\rangle \\
\text{apply } H^{\otimes n} \mapsto& (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} X^f X^w Z^{\hat{e}} \left|C_z^\perp\right\rangle \\
=& (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} (-1)^{\hat{e}\cdot w} X^f Z^{\hat{e}} X^w \left|C_z^\perp\right\rangle \\
=& (-1)^{(e+\hat{e})f} X^f Z^{\hat{e}} \left|w + C_z^\perp\right\rangle
\end{aligned}$$

2 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, at different space-time locations. Each such location records when the gate is applied and on which qubits it acts. Formally:

Definition 2.1 (Quantum Circuit)**.** A *space-time location* is a pair $l = (t, S)$, where t is a time coordinate and $S = (s_1, \ldots, s_\Delta)$ is an ordered tuple of qubit indices. For a finite collection of quantum gates

$$\mathcal{G} \subset \left\{ g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta}) \right\},$$

and integers $w, d \in \mathbb{N}$, a quantum circuit C of depth d and width w is a collection of tuples

$$C = \{u_i = (g_i, l_i)\}_i,$$

where each $g_i \in \mathcal{G}$ and each location $l_i = (l_{i,1}, l_{i,2})$ belongs to $\mathbb{Z}_d \times \mathbb{Z}_w^\Delta$. We call the tuples u_i *the gates of* C. They must satisfy the following compatibility condition: if two gates have the same time coordinate, then their supports are disjoint. Namely, for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies that the sets of qubit indices appearing in $l_{i,2}$ and $l_{j,2}$ are disjoint. We denote by

$$\mathrm{Loc}(C) := \{l_i : u_i = (g_i, l_i) \in C\}$$

the set of locations occupied by the gates of C. We define the *t-layer* of C to be its restriction to gates at time t:

$$C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}.$$

To define the action of a quantum circuit C on a given mixed state, we first associate to C a directed acyclic graph.

Definition 2.2 (Associated Computation DAG)**.** Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the directed acyclic graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set is the gate set of C, namely $\mathcal{U} = \{u_i\}_i$.
2. For $u_i = (g_i, l_i)$ and $u_j = (g_j, l_j)$, there is a directed edge $(u_i, u_j) \in E$ if and only if j is one of the minimizers of

$$\left\{ l_{k,1} : (g_k, l_k) \in \mathcal{U},\ l_{k,1} > l_{i,1},\ l_{k,2} \cap l_{i,2} \neq \emptyset \right\}.$$

   Equivalently, u_j is one of the earliest later gates whose support intersects the support of u_i. There may be more than one such minimizer, so the out-degree may exceed one.

Suppose that $J = (j_1, j_2, \ldots, j_k)$ is an ordering of the gate indices that agrees with a topological order induced by π_C. Then we define the associated partial circuits $C_J = \{C_{J,i}\}_{i=1}^{k}$ by

1. $C_{J,1} = \{u_{j_1}\}$.
2. For $1 \leq i < k$, set

$$C_{J,i+1} = C_{J,i} \cup \{u_{j_{i+1}}\}.$$

We call $C_{J,i}$ the partial computation circuit up to step i.

Definition 2.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u'_i\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

Definition 2.4 (C subjected to $\mathcal{E}$ every m computational layers)**.** Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w, let $m \in \mathbb{N}$, and set

$$d_m := \left\lceil \frac{d}{m} \right\rceil.$$

Let $\mathcal{L} \subset [d_m] \times [w]$ be a set of locations, and let $\mathcal{E}$ be a quantum circuit, such $\mathrm{Loc}(\mathcal{E}) = \mathcal{L}$. We name $\mathcal{E}$ *subsets of faults*, and we name the layers of $\mathcal{E}$ *fault layers*. The circuit C *subjected to* $\mathcal{E}$ *every* m *computational layers*, denoted by $C[\mathcal{E}]_m$, is obtained by keeping the order of the computational layers of C and inserting one fault layer after every block of m consecutive computational layers. Thus the first fault layer is inserted after the computational layers $1, \ldots, m$, the second is inserted after the computational layers $m + 1, \ldots, 2m$, and so on. Concretely, each computational gate $u_i = (g_i, (t_i, S_i))$ of C is moved to the time coordinate

$$t_i + \left\lfloor \frac{t_i - 1}{m} \right\rfloor,$$

while each fault gate $(g, (t, S)) \in \mathcal{E}$ is placed at time coordinate

$$t(m + 1).$$

Equivalently, for each time index $t \in [d_m]$, the computational layers with original times

$$(t - 1)m + 1, \ldots, \min\{tm, d\}$$

appear consecutively and are followed by the fault layer at time $b(m+1)$. When $m = 1$, the computational gates occupy the odd time coordinates and the fault layers occupy the even time coordinates. When the parameter is omitted, we mean $m = 1$. We denote the associated computation DAG of $C[\mathcal{E}]_m$ by $\pi[C, \mathcal{E}]$.

Remark 2.5. Formally, a fault circuit is just a particular quantum circuit whose gates are interpreted as faults rather than as computational operations. Thus the distinction between a quantum circuit and a fault circuit is semantic rather than structural: both are circuits in the sense of Definition 2.1, but they play different roles in the analysis.

Example 2.6. When $m = 3$, the computational layers are sent to the times

$$1, 2, 3, 5, 6, 7, 9, 10, 11, \ldots,$$

while the fault layers are sent to the times

$$4, 8, 12, \ldots.$$

The chapter uses two different notions of propagation. They are related, but they serve different analytical purposes. The first notion does not rewrite the circuit and does not change the time coordinates. Instead, it propagates part of the error inside a fixed running of the computation and asks what the accumulated effect looks like at a chosen step τ. In particular, this notion applies not only to a single fault, but also to a collection of faults propagated one after another according to the computation DAG. This is the right framework when one wants to speak about the size of the syndrome or of the deviation *at an actual intermediate time* of the computation.

This distinction is essential for statements such as the intermediate value theorem proved below. For example, a codeword may absorb noise during one time unit and be mapped to another codeword. If one only commutes whole fault layers forward, then one reorganizes the circuit globally and may lose the ability to point to the time at which a large syndrome first appears. The first notion of propagation avoids this problem: by propagating only part of the error while keeping the underlying time axis fixed, it lets us identify intermediate steps where a medium-size or large deviation must already be visible.

The second notion is more global. There we swap entire layers and then update the time coordinates accordingly. Its role is not to locate when a syndrome becomes large, but rather to compare different time-unfoldings of the same noisy circuit and to coarse-grain the noise into sparser effective layers. In particular, once the theorem proved using this second notion is established, any constant-depth collection of gates may be treated as a single time step for the purpose of the noise analysis: up to the corresponding effective rescaling of the noise rate, we may ignore the faults that occur during the internal application of those gates and replace them by an effective fault layer acting only before or after that constant-depth subcircuit. Moreover, this effective rescaling is correct with respect to local stochastic noise channels: a local stochastic noise channel is mapped to another local stochastic noise channel, with a rescaled noise rate.

For readability, we therefore separate the discussion into two subsections. The first subsection develops propagation inside a fixed running and is tailored to the deviation-cluster bounds, while the second develops layer propagation and is tailored to the coarse-graining argument.

2.1 Local Propagation Along a Running

Definition 2.7 (Fault propagation to a step τ)**.** Let $C = \{u_i\}_i$ be a quantum circuit, let $J = (j_1, \ldots, j_k)$ be an order agreeing with π_C, and let $u_i \in C$. Suppose that $u_i = u_{j_l}$ in this order. If $l > \tau$, we define the propagation of u_i to step τ to be the identity operator.

Otherwise, define operators $X_0, X_1, \ldots, X_{\tau-l}$ recursively by setting $X_0 = u_{j_l}$ and requiring that for every $1 \leq s \leq \tau - l$,

$$X_{s-1} u_{j_{l+s}} = u_{j_{l+s}} X_s.$$

We then define the *propagation of* u_i *to step* τ to be the operator

$$X := X_{\tau-l}.$$

Now let $C[\mathcal{E}]$ be a noisy circuit, let J be an order agreeing with $\pi_{C[\mathcal{E}]}$, and let τ be a step in that order. For each fault gate $e \in \mathcal{E}$ that occurs no later than step τ, denote by X_e^τ its propagation to step τ. We define the propagated error at step τ by multiplying these propagated faults in reverse topological order:

$$X^\tau := \prod_{e \preceq \tau} X_e^\tau.$$

Definition 2.8 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^τ be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 2.9. Using the same notations as in Definition 2.8, for a set of unitary operators $\mathcal{S} = \{s_i \colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}_{i=1}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

The following definitions extend the notion of a Tanner graph to the setting of quantum circuits.

Definition 2.10 (Tanner graph of a register)**.** Let C be a quantum circuit and let R be a register, namely a subset of qubits of C. The *Tanner graph* of R at step τ, denoted by $\mathcal{T}_{R,\tau}$, is defined as follows:

1. If at step τ the register R is encoded by a code $\mathcal{C}$, then $\mathcal{T}_{R,\tau}$ is the Tanner graph of $\mathcal{C}$.

2. Otherwise, $\mathcal{T}_{R,\tau}$ is the empty graph.

When the step is clear from the context, we omit the subscript τ.

Definition 2.11 (Generalized Tanner graph of a circuit)**.** Let C be a quantum circuit. For registers A and B in C, define the generalized Tanner graph of their union by

$$\mathcal{T}_{A\cup B,\tau} := \mathcal{T}_{A,\tau} \cup \mathcal{T}_{B,\tau}.$$

More generally, for a partition of the qubits into registers $\{R_i\}_i$, the generalized Tanner graph of C at step τ is

$$\mathcal{T}_{C,\{R_i\},\tau} := \bigcup_i \mathcal{T}_{R_i,\tau}.$$

When the partition and the step are clear, we abbreviate this notation to $\mathcal{T}_C$.

Definition 2.12 (Clustered deviation)**.** Let C be a quantum circuit, let R be a register of C, and let $\mathcal{E}$ be a fault set. Let $\mathcal{T}_{R,\tau}$ be the Tanner graph of R at step τ, and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_{R,\tau}|_{\mathcal{D}_\tau}$ the subgraph induced on the left vertices corresponding to the qubits in $\mathcal{D}_\tau$.

We say that two left vertices are connected if they share a common right neighbor. The *deviation clusters* at step τ are the connected components of $\mathcal{T}_{R,\tau}|_{\mathcal{D}_\tau}$. The size of a deviation cluster is the number of right vertices it contains.

Claim 2.13. Let C be a quantum circuit whose gates act on at most Δ qubits. Let R be a quantum register encoded by a quantum error-correcting code whose Tanner graph has left degree at most Δ_1 and right degree at most Δ_2. Fix a fault set $\mathcal{E}$ and an order J agreeing with $\pi_{C[\mathcal{E}]}$. For a step $\tau \in J$, let $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots$ denote the deviation clusters at step τ.

Suppose that at step τ_1 there is a deviation cluster $Y_i^{(\tau_1)}$. Then there exist deviation clusters Z and Z' at step $\tau_1 - 1$ such that

$$Z \cap Y_i^{(\tau_1)} \neq \emptyset, \qquad |Z| \geq \frac{1}{\Delta\Delta_1\Delta_2} |Y_i^{(\tau_1)}|,$$

and

$$Z' \cap Y_i^{(\tau_1)} \neq \emptyset, \qquad |Z'| \leq \Delta\Delta_1\Delta_2 |Y_i^{(\tau_1)}|.$$

Proof. Let u be the gate applied at step τ_1. Passing from step $\tau_1 - 1$ to step τ_1 can modify the deviation only on qubits touched by u. Since u acts on at most Δ qubits, and each such qubit is adjacent in the Tanner graph to at most Δ_1 checks, each of which meets at most Δ_2 qubits, the gate u can interact with vertices belonging to at most

$$\Delta\Delta_1\Delta_2$$

deviation clusters from step $\tau_1 - 1$.

Let $Y_1, \ldots, Y_m$, with $m \le \Delta\Delta_1\Delta_2$, be those deviation clusters at step $\tau_1 - 1$ that meet the neighborhood affected by u and whose union contains $Y_i^{(\tau_1)}$. Then

$$|Y_i^{(\tau_1)}| \le \sum_{j=1}^{m} |Y_j| \le m \max_{1\le j\le m} |Y_j| \le \Delta\Delta_1\Delta_2 \max_{1\le j\le m} |Y_j|.$$

Hence one of these clusters, call it Z, satisfies

$$|Z| \ge \frac{1}{\Delta\Delta_1\Delta_2} |Y_i^{(\tau_1)}|.$$

By construction, $Z \cap Y_i^{(\tau_1)} \neq \emptyset$.

For the upper bound, apply the same argument to the inverse gate $u^\dagger$, which also acts on at most Δ qubits. Viewing the transition from step τ_1 back to step $\tau_1 - 1$ as the application of $u^\dagger$, we conclude that some deviation cluster Z' at step $\tau_1 - 1$ intersecting $Y_i^{(\tau_1)}$ satisfies

$$|Z'| \le \Delta\Delta_1\Delta_2 \, |Y_i^{(\tau_1)}|.$$

This proves both bounds. □

Corollary 2.14 (Intermediate Value Theorem)**.** Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$, and let x be an integer. Assume that at the initial step τ_o, there is no deviation cluster of size more than αx, and that at a step $\tau_1 \in J$, such that $\tau_1 > \tau_o$, there is a deviation cluster $Y^{(\tau_1)}$ of size $|Y^{(\tau_1)}| > x$. Then there is a step $\tau_\star \in J$ such that $\tau_o < \tau_\star < \tau_1$ for which there is a deviation cluster Z at step $\tau_\star$, satisfying that the size of Z is in the range $(\alpha x, x)$.

Proof. Assume for contradiction that for every step τ with $\tau_o < \tau < \tau_1$, all deviation clusters have size at most αx. In particular, this holds at step $\tau_1 - 1$. On the other hand, since $|Y^{(\tau_1)}| > x$, Claim 2.13 implies the existence of a deviation cluster at step $\tau_1 - 1$ of size at least

$$\frac{1}{\Delta\Delta_1\Delta_2} |Y^{(\tau_1)}| = \alpha |Y^{(\tau_1)}| > \alpha x,$$

which is a contradiction. Therefore there must exist an intermediate step $\tau_\star$ with $\tau_o < \tau_\star < \tau_1$ at which some deviation cluster has size in the interval $(\alpha x, x)$. □

Claim 2.15. Let $\alpha = (\Delta\Delta_1\Delta_2)^{-1}$. Let C be a quantum circuit with order J and depth d, and let $\mathcal{E} \sim \mathcal{N}_\mathcal{E}$, so that $C[\mathcal{E}]$ is a random circuit whose gates act on at most Δ qubits. Fix a time $t \le 2d$ and an integer $x \in \mathbb{N}$. Let A be the event that by time t there exists a deviation cluster of size greater than x. For each step $\tau \in J$, let B_τ be the event that step τ is the first step at which there exists a deviation cluster of size in the interval $(\alpha x, x)$. Then

$$\mathbf{Pr}\big[A\big] \le 2wd \max_{\tau\in[2wd]} \mathbf{Pr}\big[B_\tau\big].$$

Proof. By Corollary 2.14, whenever the event A occurs there exists a step $\tau \in J$ with $\tau \leq 2wt$ at which there is a deviation cluster of size in the interval $(\alpha x, x)$. Hence

$$A \subset \bigcup_{\tau \in [2wt]} B_\tau \subset \bigcup_{\tau \in [2wd]} B_\tau.$$

The union bound now gives

$$\mathbf{Pr}\big[A\big] \leq \mathbf{Pr}\big[\bigcup_{\tau \in [2wd]} B_\tau\big] \leq \sum_{\tau \in [2wd]} \mathbf{Pr}\big[B_\tau\big] \leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}\big[B_\tau\big].$$

□

2.2 Layer Propagation and Time Coarse-Graining

We now switch from propagation along a fixed running to propagation at the level of entire time slices. This second notion is intentionally stronger and less local. In the previous subsection, the purpose was to say: given a fault that already occurred, what is its effect at a later step of the same computation? Here the purpose is different: we want to reorganize the circuit itself, by commuting fault layers forward and grouping them into coarser effective layers. This is the mechanism behind the later renormalization claim for local stochastic noise.

In particular, local propagation keeps the ambient circuit fixed and changes the *description of the error*. Layer propagation, by contrast, changes the *presentation of the circuit* while preserving the implemented channel. This is why the two notions should not be merged into one definition: one is adapted to tracking deviations, while the other is adapted to comparing circuits related by time-unfolding.

Definition 2.16 (Layer propagation)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit. For each time coordinate t, write $U_t := C|_t$. The family of non-empty t-cuts $\{U_t\}_t$ is called the *layer decomposition* of C.

Let $t < t'$ be such that U_t and $U_{t'}$ are non-empty and $U_s = \emptyset$ for every $t < s < t'$. The *forward propagation* of the layer U_t across the next layer $U_{t'}$ is the operation that replaces the pair $(U_t, U_{t'})$ by a pair $(U_{t'}, V_{t\to t'})$ satisfying

$$\prod_{u \in U_t} u \prod_{v \in U_{t'}} v = \prod_{v \in U_{t'}} v \prod_{w \in V_{t\to t'}} w.$$

The layer $V_{t\to t'}$ is called the *propagation expression* of U_t across $U_{t'}$.

More generally, if $t < t'$, the *forward propagation of U_t to time t'* is obtained by iterating the above operation through the successive non-empty cuts between times t and t'. The resulting propagated layer is denoted by $V_{t\to t'}$ and is again called the propagation expression of U_t to time t'. The resulting circuit has layer decomposition

$$\ldots, U_{t-1}, U_t, U_{t+1}, \ldots, U_{t'}, U_{t'+1}, \ldots \mapsto \ldots, U_{t-1}, U_{t+1}, \ldots, U_{t'}, V_{t\to t'}, U_{t'+1}, \ldots.$$

After propagation we update the time coordinates so that they agree with the new layer in which each gate is applied.

Claim 2.17. Assume that every gate in C acts on at most Δ qubits. Then every circuit obtained from C by forward layer propagation also consists of gates acting on at most Δ qubits.

Proof. Propagating a gate $u \in U_i$ across a layer U_{i+1} conjugates u by the product of the gates in U_{i+1} that intersect its support. Since the gates inside U_{i+1} are pairwise disjoint, each qubit in the support of u participates in at most one gate from U_{i+1}. Therefore the support of the propagated image of u is contained in the union of the supports of at most Δ gates, each of arity at most Δ, but only one such gate can be associated with each qubit of u. Consequently the propagated image still acts on at most Δ wires. Applying this argument repeatedly proves the claim. □

Definition 2.18 ($C \sim_P C'$)**.** For quantum circuits C and C', we write $C \sim_P C'$ if one can be obtained from the other by a finite sequence of layer propagations.

By construction, $\sim_P$ preserves the implemented channel. Indeed, if $C \sim_P C'$ and ρ is any mixed state on the input register, then

$$C \circ \rho = C' \circ \rho.$$

Definition 2.19 (Noise channel)**.** A *noise channel* $\mathcal{N}_\mathcal{E}$ is a distribution over sets of fault locations.

Definition 2.20 (Local stochastic noise channel)**.** Let $p \in (0, 1)$. We say that the noise channel $\mathcal{N}_\mathcal{E}$ is *local stochastic with rate* p if for every finite set of locations $S \subset \mathbb{N} \times \mathbb{N}$,

$$\mathbf{Pr}_{\mathcal{E} \sim \mathcal{N}_\mathcal{E}} \big[S \subset \mathrm{Loc}(\mathcal{E}) \big] \leq p^{|S|}.$$

Given a circuit C and a random fault set $\mathcal{E} \sim \mathcal{N}_\mathcal{E}$, the subjected circuit $C[\mathcal{E}]$ is itself a random variable.

Definition 2.21 (Witness family in the backward light cone)**.** Let W be a propagation expression obtained from a block of c layers of a noisy circuit after propagating all even fault layers to the end of the block. For a subset $S \subset \mathrm{Loc}(W)$, we define the *witness family* of S, denoted by $\mathcal{W}(S)$, to be the collection of all minimal sets

$$S' \subset \Lambda(S)$$

such that there exists a fault pattern containing S' whose propagation produces all the faults at the locations in S. Here minimality is with respect to inclusion.

Claim 2.22. Let $c \geq 2$ be even, let C be a depth-$c/2$ circuit whose gates have arity at most Δ, and let $C[\mathcal{E}]$ be the circuit obtained by inserting the faults of $\mathcal{E}$ between the computational layers. Write the layer decomposition of $C[\mathcal{E}]$ as

$$U_0, U_1, \ldots, U_c,$$

where the even layers are fault layers and the odd layers are computational layers. Propagate the fault layers $U_2, U_4, \ldots, U_{c-2}$ forward beyond U_{c-1}, in reverse order, and denote by $V^{(1)}$ the resulting propagation expression obtained by grouping together those propagated layers with the terminal fault layer U_c. Then for every $S \subset \mathrm{Loc}(V^{(1)})$ the witness family $\mathcal{W}(S)$ is non-empty, and every element $S' \in \mathcal{W}(S)$ satisfies

$$S' \subset \Lambda(S) \cap \bigcup_{k=0}^{c/2} U_{2k}$$

and

$$|S'| \geq \frac{|S|}{\Delta^c}.$$

Consequently,

$$\{S \subset \mathrm{Loc}(V^{(1)})\} \subset \bigcup_{S' \in \mathcal{W}(S)} \{S' \subset \mathrm{Loc}(\mathcal{E})\}.$$

Proof. Suppose that the event $\{S \subset \mathrm{Loc}(V^{(1)})\}$ occurs. Then there exists at least one set of original fault locations whose propagation produces all the faults in S. Among all such sets choose one that is minimal with respect to inclusion, and denote it by S'. By construction, $S' \in \mathcal{W}(S)$, so the witness family is non-empty.

Since every element of S is obtained by propagating some original fault through at most c layers, and propagation never leaves the backward light cone of the produced qubits, every location in S' lies in $\Lambda(S) \cap \bigcup_{k=0}^{c/2} U_{2k}$.

It remains to bound the size of S'. A single fault location can influence at most Δ^c qubits in the final layer of the block, because at each of the at most c propagation steps the support can branch to at most Δ qubits. Therefore any set of original faults capable of generating all elements of S must contain at least $|S|/\Delta^c$ fault locations. In particular, this holds for the minimal producing set S', so $|S'| \geq |S|/\Delta^c$.

Finally, whenever $S \subset \mathrm{Loc}(V^{(1)})$ occurs, at least one minimal producing set $S' \in \mathcal{W}(S)$ is fully contained in the realized fault set $\mathrm{Loc}(\mathcal{E})$. This proves the event inclusion. □

Claim 2.23. Let $\mathcal{N}_{\mathcal{E}}$ be a local stochastic noise channel with rate p, and let C be a quantum circuit whose gates have arity at most Δ. Fix an even integer c. Then there exists a local stochastic noise channel $\mathcal{N}_{\mathcal{E}_q}$ with rate

$$q := \left(2^{H(\Delta^{-c})\Delta^c} p\right)^{\Delta^{-c}},$$

such that

$$C[\mathcal{E}]_1 \sim_P C[\mathcal{E}_q]_{c/2}.$$

In particular, up to a constant depending only on Δ and c, the rate q behaves like $p^{\Delta^{-c}}$.

Proof. Partition the subjected circuit $C[\mathcal{E}]_1$ into consecutive blocks of c layers. Inside each block, propagate all internal even fault layers to the end of the block, exactly as in Claim 2.22. Doing this block by block yields a new circuit C' in which each block of $c/2$ computational layers is followed by a single coarse-grained fault layer. Therefore $C' = C[\mathcal{E}']_{c/2}$ for a suitable random fault set $\mathcal{E}'$, and by construction

$$C[\mathcal{E}]_1 \sim_P C[\mathcal{E}']_{c/2}.$$

It remains to prove that $\mathcal{E}'$ is local stochastic. Fix one coarse-grained propagation expression $V^{(i)}$ and a subset $S \subset \mathrm{Loc}(V^{(i)})$. By Claim 2.22, the event $\{S \subset \mathrm{Loc}(V^{(i)})\}$ implies that at least one witness set $S' \in \mathcal{W}(S)$ is fully contained in the original fault set $\mathrm{Loc}(\mathcal{E})$, and every such witness satisfies $|S'| \geq |S|/\Delta^c$. Hence

$$\begin{aligned}\mathbf{Pr}\big[S \subset \mathrm{Loc}(V^{(i)})\big] &\leq \mathbf{Pr}\Big[\bigcup_{S' \in \mathcal{W}(S)} \{S' \subset \mathrm{Loc}(\mathcal{E})\}\Big] \\ &\leq \sum_{S' \in \mathcal{W}(S)} \mathbf{Pr}\big[S' \subset \mathrm{Loc}(\mathcal{E})\big].\end{aligned}$$

The channel $\mathcal{N}_{\mathcal{E}}$ is local stochastic with rate p, so each term is at most $p^{|S'|}$. Moreover, the backward light cone of each effective fault has size bounded by a constant depending only on Δ and c, so the number of candidate witnesses of size ℓ is at most

$$\binom{\Delta^c |S|}{\ell}.$$

Choosing $\ell = \lceil |S|/\Delta^c \rceil$ and using the standard entropy bound

$$\binom{\Delta^c |S|}{|S|/\Delta^c} \leq 2^{H(\Delta^{-2c})\Delta^c |S|},$$

we obtain

$$\mathbf{Pr}\big[S \subset \mathrm{Loc}(V^{(i)})\big] \leq \Big(2^{H(\Delta^{-2c})\Delta^c} p\Big)^{|S|/\Delta^c} = q^{|S|}.$$

Since this bound is uniform over all subsets S and all coarse-grained layers, the induced random set $\mathcal{E}'$ is local stochastic with rate q. Setting $\mathcal{E}_q = \mathcal{E}'$ completes the proof. □

3 The (d, d')-Dimensional Toric Code

In this section, the parameter d denotes the geometric dimension of the toric complex. This use of the letter d is local to the present section.

Definition 3.1 (d-Dimensional Toric Complex)**.** For integers d and n, denote the group resulting from the d-directed power of the cyclic group of order n by $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$, and denote the standard generating set of $\mathcal{G}_n^d$ by $S = \{1_i \colon i \in [d]\}$. We define the *$d$-dimensional toric complex* to be the hypergraph obtained from the Cayley graph

$\mathrm{Cay}(\mathcal{G}_n^d, S)$ by taking its vertices as the 0-faces and for any $i \in [d-1]$ we define the i-faces as follow: First, for a vertex $v \in \mathcal{G}_n^d$ and for a subset of generators $S' \subset S$ of size $|S'| = i$, the i-face rooted at v and spanned by S', denoted by $\theta_{v,S'}$, is:

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\}$$

The i-faces of the complex are the collection of all the possible rooted spanned i-faces induced by $\mathrm{Cay}(\mathcal{G}_n^d)$. We will denote them by $\mathcal{F}_i$.

Additionally, we name n and d, the *side length* and the *dimension* of the complex.

Definition 3.2 (Positive Edges and Cubes)**.** For $u, v \in \mathcal{G}_n^d$, and $S_1', S_2' \subset [d]$, with sizes $|S_1'| = |S_2'| = i$, we say that the i-faces $\theta_{v,S_1'}$ and $\theta_{u,S_2'}$ are adjacent if the intersection $\theta_{v,S_1'} \cap \theta_{u,S_2'}$ is an $i-1$-face. In addition, if θ_1 and θ_2 are i and $i-1$-faces such that $\theta_2 \subset \theta_1$, then we say that θ_2 is a *positive edge relative to face* θ_1 if they are rooted at the same vertex. Furthermore, we say that θ_1 is a *positive cube* of θ_2. When the i-face is clear from the context, we will abbreviate and say that θ_2 is *positive*.

Example 3.3. For example, let $v, u \in \mathcal{G}_n^d$ and $g_1 \neq g_2 \in S$. Consider the 2-face $\theta = \{v, g_1 v, g_2 g_1 v, g_2 v\}$. Then, for $x, y \in \mathcal{G}_n^d$, we say that an edge (1-face) $\{x, y\}$ is *positive relative to the face* θ if $x = v$ and y is either $g_1 v$ or $g_2 v$.

We denote the complex by $G = (V, E, F)$, where V, E, and F are the 0, 1, and 2 faces.

Although the present paper is not organized in the language of chain complexes, the boundary operators below are the standard ones for toric codes. We use them because they give a compact description of the local constraints and make the CSS structure transparent. We do not need the full homological formalism here, but it remains the natural background framework for the code family.

Definition 3.4 (Assignments on a toric complex)**.** Let $\mathcal{T}$ be a d-dimensional toric complex. For each i, the set of i-faces induces a vector space $\mathbb{F}_2^{|\mathcal{F}_i|}$. An element $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$ is called an *assignment on the i-faces* of $\mathcal{T}$.

For a vertex v and a subset of generators $S' \subset S$ of size $|S'| = i$, let $\theta_{\mathbf{v},\mathbf{S'}}$ denote the standard basis vector corresponding to the i-face $\theta_{v,S'}$. Thus the family

$$\left\{ \theta_{\mathbf{v},\mathbf{S'}} : v \in \mathcal{F}_0,\ S' \subset S,\ |S'| = i \right\}$$

is the standard coordinate basis of $\mathbb{F}_2^{|\mathcal{F}_i|}$.

Let z be an assignment on the i-faces. Let $v_1, \ldots, v_k \in \mathcal{F}_0$ and $S_1', \ldots, S_k' \subset S$ be such that $\theta_{v_1,S_1'}, \theta_{v_2,S_2'}, \ldots, \theta_{v_k,S_k'}$ are precisely the i-faces in the support of z, namely:

$$z = \sum_{i=1}^{k} \theta_{\mathbf{v_i},\mathbf{S_i'}}$$

We refer to $\theta_{v_1,S'_1}, \theta_{v_2,S'_2}, \ldots, \theta_{v_k,S'_k}$ as the collection of faces that form z, or briefly, the faces form z. For $i > 0$, we define the linear map $\partial : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ such that

$$\partial\theta_{\mathbf{v},\mathbf{S}'} := \sum_{g\in S'} \theta_{\mathbf{v},\mathbf{S}'/\mathbf{g}} + \sum_{g\in S'} \theta_{\mathbf{gv},\mathbf{S}'/\mathbf{g}}$$

For $i < d$, we define the linear map $\partial^* : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ such that

$$\partial^*\theta_{\mathbf{v},\mathbf{S}'} := \sum_{g\in S/S'} \theta_{\mathbf{v},\mathbf{S}'\cup\mathbf{g}} + \sum_{g\in S'} \theta_{\mathbf{g^{-1}v},\mathbf{S}'\cup\mathbf{g}}$$

We define the *restriction of z to a vertex v* to be the linear map $|_v \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S}'}|_v := \begin{cases} \theta_{\mathbf{u},\mathbf{S}'} & v \in \theta_{u,S'} \\ 0 & \text{else} \end{cases}$$

We define the *front of z to a vertex v* to be the linear map $|_{v+} \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S}'}|_{v+} := \begin{cases} \theta_{\mathbf{u},\mathbf{S}'} & u = v \\ 0 & \text{else} \end{cases}$$

We say that an assignment z is *connected* if the graph that is induced by taking the faces that support z as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of z to be its subsets such that each of them is a connected assignment. For a j-face f, we say that $f \in z$ if z has a support on $\theta_{\mathbf{v},\mathbf{S}'}$ such that $f \in \theta_{v,S'}$. Consider an assignment $z \in \mathbb{F}_2^{\mathcal{F}_i}$ and a vertex $v \in \mathcal{G}_n^d$. We say that z is a *rooted assignment* at vertex v if there are subsets $S'_1, \ldots, S'_k \subset S^{\times k}$ such that each of S'_j is at size i and z is decomposed as:

$$z = \sum_{j}^{k} \theta_{\mathbf{v},\mathbf{S}'_\mathbf{j}}$$

Put differently, z is the result of taking a collection of i-faces that are rooted at v.

Claim 3.5 (CSS structure)**.** For every i, the maps $\partial \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ and $\partial^* \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ satisfy the usual orthogonality relation between boundary and coboundary operators. Consequently, the X-checks defined by ∂ and the Z-checks defined by ∂^* commute, and therefore the resulting toric code is a CSS code.

Proof. It is enough to check the claim on basis vectors. For a basis vector $\theta_{\mathbf{v},\mathbf{S}'}$, the appendix calculation shows that

$$\partial\big(\partial\theta_{\mathbf{v},\mathbf{S}'}\big) = 0.$$

The same cancellation argument applies dually to the coboundary operator. Equivalently, every $d'-1$ face and every $d'+1$ face overlap on an even number of d'-faces, so the corresponding X- and Z-checks commute. This is exactly the CSS condition. □

We will use only this CSS property and the local geometry of the code; standard facts about distance and rate may be found in the toric-code literature, for example in [Den+02].

Definition 3.6 ((d, d') Toric Code)**.** Let d, d' be integers such that $1 < d' < d$. For any integer n, we construct the nth instance of the quantum code family (d, d') *Toric Code* as follows: Let $G = (\mathcal{F}_0, \ldots, \mathcal{F}_d)$ be the d-dimensional toric complex obtained from $\mathcal{G}_n^d$ as defined in Definition 3.1. We associate the (q)bits with the $\mathcal{F}_{d'}$, the X-checks with the $\mathcal{F}_{d'-1}$, and the Z-checks with the $\mathcal{F}_{d'+1}$. An X-check, c_x, is satisfied if the parity of (q)bits set on the d'-face containing c_x is even. Similarly, a Z-check, c_z, is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the d'-faces contained in c_z is even.

Claim 3.7. For any generator subset S' of size d', denote by $x_{S'}$ and $z_{S'}$ the following assignments:

$$x_{S'} := \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'} \quad z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u},\mathbf{S}'}$$

Then $x_{S'}$ and $z_{S'}$ are binary representations of anticommuting logical codewords of the (d, d')-Toric code. In particular, $x_{S'} \in \mathcal{C}_X/\mathcal{C}_Z^\perp$, $z_{S'} \in \mathcal{C}_Z/\mathcal{C}_X^\perp$, and $x_{S'} z_{S'} = 1$. In addition, let $X_{S'}$ and $Z_{S'}$ be the Pauli matrices obtained by multiplying each non-trivial coordinate of them by X or Z, respectively.

Proof. By definition $x_{S'}$ and $z_{S'}$ are binary assignments on the d'-dimensional faces. We start by showing that $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$:

$$\begin{aligned}
\partial^- \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{gu},\mathbf{S}'/\mathbf{g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{gu},\mathbf{S}'/\mathbf{g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} = 0
\end{aligned}$$

$$\begin{aligned}
\partial^+ \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} + \theta_{\mathbf{g}^{-1}\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}^{-1}\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}' \cup \mathbf{g}} = 0
\end{aligned}$$

Thus $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabiliers, it's enough to show they admit no vanishing product.

$$x_{S'} z_{S'} = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'} \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u},\mathbf{S}'} = \theta_{\mathbf{e},\mathbf{S}'} \theta_{\mathbf{e},\mathbf{S}'} = 1$$

□

Definition 3.8 (The Transpose Map)**.** Let d, d' be integers, and let $\mathcal{G}$ be a (d, d')-Toric complex generated by the generator set S. We define the *transpose map* $\phi : \mathcal{F}^{d'} \to \mathcal{F}^{d-d'}$ by acting on the basis elements as follows:

$$\phi(\theta_{\mathbf{v},\mathbf{S}'}) \to \theta_{\mathbf{v}^{-1},\mathbf{S}/\mathbf{S}'}$$

Notice that for $d' = d - d'$, the transpose map is a permutation over the bits.

Definition 3.9 (Hadamard-Type Gate H_ϕ)**.** For $d' = d - d'$, we define the logical Hadamard-type gate H_ϕ: $L \to L$ as follows, for any $S' \subset S$, of size d':

$$H_\phi(X_{S'}) \to Z_{S/S'}$$
$$H_\phi(Z_{S'}) \to X_{S/S'}$$

[COMMENT] Change the definition above such that it will be about moving to the dual code $C_X/C_Z^\perp \to C_Z/C_X^\perp$. And not about the logical Hadamard.

Claim 3.10. The Hadamard-type gate H_ϕ has a fold-transversal implementation by, first applying H^n over all the physical qubits, and then permuting them by the transpose map ϕ, namely $\hat{H}_\phi = \phi \circ H^n$.

Proof. Clearly, for any subset of generators $S' \subset S$, the map ϕ sends $x_{S'}$ to $z_{S/S'}$:

$$\phi(x_{S'}) = \phi(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'}) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'} = z_{S/S'}$$

Moreover, ϕ sends binary representations of Z-stabilizers to binary representations of X-stabilizers, and vice versa. To see this, consider a $d'+1$ dimensional face, and consider the image of the stabilizer it defines.

$$\begin{aligned}
\phi\left(\partial^- \theta_{\mathbf{v},\mathbf{S}'}\right) &= \phi\left(\sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{gv},\mathbf{S}'/\mathbf{g}}\right) \\
&= \sum_{g \in S'} \theta_{\mathbf{v}^{-1},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}} + \theta_{\mathbf{g}^{-1}\mathbf{v}^{-1},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}} \\
&= \partial^+ \theta_{\mathbf{v}^{-1},\mathbf{S}/\mathbf{S}'}
\end{aligned}$$

And since $\phi \circ \phi = I$, we also have that $\phi\left(\partial^+ \theta_{\mathbf{v}^{-1},\mathbf{S}/\mathbf{S}'}\right)$. Thus, $H^{\otimes n} \circ \phi$ sends Z-stabilizers to X-stabilizers and vice versa, thereby preserving the code space. Combining this with the above, $H^{\otimes n} \circ \phi$ realizes the logical form H_ϕ in the $(d, d/2)$-Toric code. □

[COMMENT] Here, add that the measurement in the X-basis can be done using $\mathbf{H}^n$, measurement, and then $\log(n)$ classical depth.

[COMMENT] Add a corollary that state that we can perform a gate teleportations of $\mathbf{S}$ and $\mathbf{H}$, with the resource states $|\mathbf{S}\rangle$ and $|\mathbf{H}\rangle$, $\log(n)$ classical depth, and only transversal operation over the encoded register.

Definition 3.11 (The Nice Encoding)**.** Consider $(d, d/2)$-Toric code and fix a subset $S' \subset S$, at size d'. We define the *nice encoding* to be the subsystem code, which uses $x_{S'}$ to encode a single logical qubit. Namely, the logical codewords are:

$$|\bar{0}\rangle = \left|\mathcal{C}_Z^{\perp}\right\rangle$$
$$|\bar{1}\rangle = \left|x_{S'} + \mathcal{C}_Z^{\perp}\right\rangle$$

Definition 3.12 (Decoding Gadget)**.** Let C be a quantum circuit, and let u be a gate in C with fan-in at most Δ. We say that u is a *decoding gadget* if it has the following form:

1. The wires of u wires can be decomposed into two subsets A and B such that the wires in A are reset wires, and there exists an encoded register R in C such that the wires in B are all set in R.

2. For a set of faults $\mathcal{E}$, there is a Pauli gate u' that acts only on B such that the action of u in $C[\mathcal{E}]$ equals u'. Put differently, the quantum circuit obtained from $C[\mathcal{E}]$ by replacing u with u' equals $C[\mathcal{E}]$.

Definition 3.13 (Unitary Decoding Representation)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit, and let $\mathcal{U} = \{u_i'\}_{i=1} \subset C$ be a subset of the gates in C that are decoding gadgets. For a set of faults $\mathcal{E}$, we define the *unitary decoding representation* of $C[\mathcal{E}]$ to be the quantum circuit obtained by replacing each of the gates in $\mathcal{U}$ with their corresponding Pauli gate according to $\mathcal{E}$.

Definition 3.14 (Register coordinate system)**.** Let $C = \{u_i\}_i$ be a quantum circuit of width w with registers $\mathcal{R} = R_1, \ldots, R_k$, each of length at most n. For each register R_i, define its identifier by $\mathbf{1}_{R_i} = \mathbf{1}_i$. We define the map

$$\phi_{\mathcal{R}} \colon \mathbb{Z}_w \to \mathbb{Z}_R \times \mathbb{Z}_n$$

by setting

$$\phi_{\mathcal{R}}(x) = (\mathbf{1}_{R_j}, x'),$$

where R_j is the register containing x, and x' is the position of that qubit within R_j.

The presentation of C in the *register coordinate system* is obtained by replacing every spatial coordinate by its image under $\phi_{\mathcal{R}}$. Thus, if $u_i = (g_i, l_i) \in C$ and the space-location part of l_i is $l_{i,2} = (l_{i,2}^{(1)}, \ldots, l_{i,2}^{(\Delta)})$, then we replace it by

$$(\phi_{\mathcal{R}}(l_{i,2}^{(1)}), \ldots, \phi_{\mathcal{R}}(l_{i,2}^{(\Delta)})).$$

We call the first coordinate of $\phi_{\mathcal{R}}(x)$ the *register coordinate* and the second the *inner coordinate*.

4 Infinite Torics

Definition 4.1 (Shifting in Registers System)**.** Consider the register coordinate system induced by a quantum circuit of width w, with R registers, each of length at most n. For an integer $r \in \mathbb{Z}$, we define the r-shifting function $\mathrm{Sh}_r \colon (\mathbb{Z}_R \times \mathbb{Z}_n)^{\times \Delta} \to (\mathbb{Z}_R \times \mathbb{Z})^{\times \Delta}$ to act on space-location coordinates as follows:

$$\mathrm{Sh}_r(l_1, .., l_k) = (\mathrm{Sh}_r(l_1), .., \mathrm{Sh}_r(l_k))$$
$$\text{Where } \mathrm{Sh}_r(l_i) = \begin{cases} l_i + r & \text{if } l_i \text{ is an inner coordinate} \\ l_i & \text{else} \end{cases}$$

Definition 4.2. Let n be an integer, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with registers $\mathcal{R} = R_1, \ldots, R_k$ such that each register R_i is a toric encoding with a side length n. In addition, all the decoding gadgets of C are sweep rule gadgets; let us denote them by $\mathcal{U}$. We define the ∞-*extension of* C, denoted by C^∞, as follows:

1. For a register $R_i \in \mathcal{R}$, denote by d_i, d_i' the dimension of the toric complex that induces the code, and the dimension of the faces that are associated with the bits. Each register R_i in $\mathcal{R}$ is mapped to a register R_i', which is a (d_i, d_i')-dimensional, infinite side length, toric encoding. Since the mapping between registers is one-to-one, we use the identifier $\mathbf{1}_{R_i}$ to identify also R_i'.

2. Each gate $u_i \in C/\mathcal{U}$ is mapped to:

$$u_i = (g_i, l_i) \mapsto \bigcup_{j=-\infty}^{\infty} \{(g_i, (\mathrm{Sh}_{jn}(l_i))\}$$

3. Each gate $u \in \mathcal{U}$ is mapped to the collection of sweep rule gadgets $\{v_j\}_{j=-\infty}^{\infty}$ such that each of them has the same time coordinate as u, is in the register that corresponds to the image of u's register, and is rooted at $\mathrm{root}(v) + jn$.

For a set of faults $\mathcal{E} = \{f_i\}_{i=1}$, we denote by $C^\infty[\mathcal{E}^\infty]$ the ∞-*extension of* $C[\mathcal{E}]$.

Claim 4.3. Let $u = (g, l)$, and let $v_0, v_{\pm 1}$ be its translates in the ∞-extension rooted at $\mathrm{root}(u)$ and $\mathrm{root}(u) \pm n$. Then

$$v_0 v_{\pm 1}|_{[w]} = u.$$

Proof. The gates v_0 and $v_{\pm 1}$ are copies of the same local Pauli operator acting on translates of the same finite pattern. When restricted to the fundamental domain $[w]$, all factors outside $[w]$ disappear, and the remaining action coincides with the original gate u. Thus the product of the relevant translates restricts exactly to u. □

Definition 4.4. For a quantum circuit C, of the form defined in Definition 4.2, and a set of faults $\mathcal{E}$, denote:

$$\rho := C[\mathcal{E}] \, |0\rangle\langle 0| \, C[\mathcal{E}]^\dagger$$
$$\rho_\infty := C^\infty[\mathcal{E}^\infty] \, |0\rangle\langle 0| \, (C^\infty[\mathcal{E}^\infty])^\dagger$$

We say that a quantum circuit C is ∞-*compatible* if, for any set of faults $\mathcal{E} = \{f_i\}_{i=1}$, it holds that:

$$\rho = \mathbf{Tr}_{/[w]}\left(\rho_\infty\right)$$

Claim 4.5. Let C be quantum circuit of the form defined in Definition 4.2. Then C is ∞-compatible.

Proof. It is enough to prove the claim for circuits in their unitary decoding representation. We prove it by induction on the number of the gates.

1. Base. For an empty circuit, ρ_∞ is just the infinite string of zeros; thus, the restriction of it to $[w]$ locations, namely $\mathbf{Tr}_{/[w]}(\rho_\infty)$, yields a single matrix element corresponds to the w-length string of zeros, which is exactly ρ.

2. Step. Assume the claim holds for every circuit with fewer than τ gates, and consider a circuit C with τ gates. Denote by C' the circuit obtained from C by erasing its last gate; then C' has $\tau - 1$ gates. Denote by ρ' and ρ'_∞ the output states of $C'[\mathcal{E}]$ and $(C')^\infty[\mathcal{E}^\infty]$, respectively. By the induction assumption we have:

   $$\rho' = \mathbf{Tr}_{/[w]}\left(\rho'_\infty\right)$$

   Denote the last gate in C by u, and by $I = \{v_j\}_{-\infty}^{\infty}$ the collection of gates in C^∞ to which u is mapped. Let us split into cases, depending on whether u is or is not a decoding gate.

   (a) Suppose that u is not a decoding gate. Let g and l be the gate element and the location of u. By the definition of ∞-extension, I equals:

   $$I = \bigcup_{j=-\infty}^{\infty} \{(g, (\mathrm{Sh}_{jn}(l))\}$$

   So there is only one gate in I, corresponding to $j = 0$, that has support on qubits with spatial location in the range $[w]$. Denote that gate by v_0.

$$\begin{aligned}
\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) &= \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty}^{\infty} v_j \rho'_\infty \prod_{j=-\infty}^{\infty} v_j^\dagger\right) \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty \prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger \prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\rho'_\infty\right) v_0^\dagger \\
&= v_0 \rho' v_0^\dagger \\
&= \rho
\end{aligned}$$

(b) In the case that u is a decoding gate, observe that there are at most two gates in I that have support on the qubits with space-location in $[w]$. If there is exactly one, then repeating the non-decoding gate case while setting this gate as v_0 gives the desired result. Else, in the case where two gates act non-trivially on qubits at $[w]$, denote those gates by v_0, v_1. Since we assumed that C is given in its unitary decoding representation, it holds that v_0, v_1 are Paulis, in particular a product operator. Denote their product decompositions by $v_0 = \prod_k v_0^{(k)}$ and $v_1 = \prod_k v_1^{(k)}$. Thus, by repeating the non-decoding case, but extracting from the infinite product only the terms in v_0 and v_1 that act non-trivially at $[w]$, we get:

$$\mathbf{Tr}_{/[w]}(\rho_\infty) = v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger$$

Moreover, since v_0 and v_1 are translates of u in locations $\mathrm{root}(u)$ and $\mathrm{root}(u) \pm n$, then by Claim 4.3, we have that $u = v_0 v_1|_{[w]}$; therefore:

$$\begin{aligned}\mathbf{Tr}_{/[w]}(\rho_\infty) &= v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger \\ &= u \rho' u^\dagger \\ &= \rho\end{aligned}$$

□

The computation DAG lifts naturally to the ∞-extension: gates that arise as translates of the same local gate act on disjoint translated supports inside different periods, so they may be executed simultaneously. The same periodic lifting applies to the underlying geometric objects, passing from the finite toric complex to its infinite cover.

Definition 4.6 (The standart embedding)**.** Consider the d-dimensional infinite grid $\mathcal{G}^d$. The *standard embedding* $\mathcal{G}^d \to \mathbb{R}^d$ is given by fixing a vertex in $\mathcal{G}^d$ to be sent to the origin and then mapping the generators of the grid to the standard basis. We extend notation as follows: for a vertex $v \in \mathcal{G}^d$, a generator $g \in S$, and a coordinate $i \in [d]$, such that under the embedding $g \to i$, we denote by $v_i = v_g$ the i'th coordinate of the image of v.

Definition 4.7 (Potential Walls)**.** We define the *potential walls* to be the $d+1$ functions $L_1, L_2, .., L_{d+1}$, from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}^d \to \mathbb{R}$ as $L_i(v) := v_i$. In addition, for the generator $g \in S$ which is mapped to i, i.e $g \to i$, we identify $L_g = L_i$.
2. We define $L_{d+1} \colon \mathcal{G}^d \to \mathbb{R}$ by $L_{d+1}(v) := \sum_{i=1}^d v_i$.

Consider an assignment z over the faces. For a vertex $v \in z$, and a potential wall $l \in \{L_1, .., L_{d+1}\}$, we say that v is a *local maximum* of l in z if for any neighbor u of v, such that $u \in z$, we have $l(v) \geq l(u)$. We say that v is a *strong local maximum* if the inequality is strict, namely $l(v) > l(u)$. When the assignment z is clear from the context, we might omit it and briefly say that v is a local maximum of l.

With the assistance of potential walls, we can discuss the endpoint of a bunch of bits, which will soon become error clusters.

5 SWEEP Rule

Definition 5.1 (Sweep Rule)**.** Let z be an assignment on the d'-faces. The *sweep rule decoding procedure* at a vertex $v \in \mathcal{G}_n^d$ is defined as follows:

Measure the checks rooted at v and look for a rooted assignment $\tilde{z}$ at v such that $(\partial\tilde{z})|_{v+} = \partial(z)|_{v+}$. In other words, the X-checks rooted at v have the same syndromes for z and $\tilde{z}$. If such an assignment exists, flip $\tilde{z}$.

Definition 5.2 (Sweep cycle)**.** We define the *sweep cycle* to be the following procedure:

1. For each vertex, perform the sweep rule.
2. Apply the Hadamard transform $H^{\otimes n}$ followed by the ϕ-transpose, and flip the positive direction.
3. Again, apply the sweep rule at each vertex.
4. Reverse stage 2 by reordering the qubits according to the ϕ-transpose, and then apply the Hadamard transform $H^{\otimes n}$. Flip the positive direction again.

Claim 5.3. The local gates that perform the SWEEP rule are decoding gadgets.

Proof. A local SWEEP update acts on a constant-size neighborhood around a vertex, uses only local syndrome information, and may be implemented with fresh ancillas that are reset after the update. Any dependence on faults inside the gadget can therefore be pushed onto the data wires as an effective Pauli operator supported on the encoded register, while the reset ancillas carry no residual information. This is exactly the form required in the definition of a decoding gadget. □

6 Shrinking

The following two claims relate the local maximum points of $L_1, \ldots, L_d$ in an assignment z to the structure of the syndrome they observe. Later, we will use that relation to infer that the decoding step of the SWEEP rule can't spread the error to a vertex[1] with a higher value than the current global maximum of $L_1, \ldots, L_d$. In other words, the error does not spread in the positive directions.

Claim 6.1. Let z be an assignment, and recall that the vertices in ∂z are the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a generator in S such that v is a local maximum of L_g in ∂z. Then $gv \notin \partial z$.

Proof. Suppose $gv \in \partial z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂z. □

[1] To faces whose vertices have higher values of $L_1, \ldots, L_d$ than the current global maximum values.

Claim 6.2. Let z be an assignment, let v be a vertex in ∂z, and let g be a generator in S such that v is a local maximum of L_g. Then, for any check associated with a $d'-1$ face rooted at v that contains gv, it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d'-1$ positive face rooted at v, which contains gv. This implies $gv \in \partial z$, contradicting Claim 6.1. □

For the $(d+1)$th potential wall, we would like to have a stronger claim that implies the error cluster shrinks. For this, we start by showing that any error is equivalent, up to stabilizer addition, to an error whose left-corner edge sees a syndrome. We present two notations of local codes that describe the local view of a $(d+1)$th edge that sees no syndrome. They are the *stabilizers code at vertex* v and the *small code at vertex* v. The first corresponds to stabilizers rooted at the vertex, while the second is the collection of local views on its positive faces that satisfy its positive checks. Then, in Claim 6.5, we consider a vertex which is a $(d+1)$th edge and doesn't see a syndrome, meaning that its local view is in its *small code*, and show that there is a codeword of its *stabilizers code* that agrees with its positive faces. Addition by that stabilizer yields a new assignment excluding v.

Definition 6.3 (Stabilizers Code at Vertex v)**.** We define *stabilizer code at vertex* v, denoted by Stab_v, as the restriction of the (d,d')-Toric code to the d'-dimensional faces such that their vertices are at a distance of at most one positive step in each direction from v. That is equivalent to the union of all the positive d'-dimensional faces of v with the d-dimensional faces of each of its sibling gv, excluding faces containing g as their generator. Namely:

$$\begin{aligned}\mathrm{Stab}_v := \big\{ z : \partial^- z = 0 \text{ and } \\ z \in \mathrm{Span}\big(\big\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d'\big\} \\ \bigcup_{g\in S} \big\{\theta_{\mathbf{gv},\mathbf{S'/g}} : S' \subset S/g, |S'| = d'\big\}\big)\big\}\end{aligned}$$

The checks of the code, are of the following three types:

1. Checks that are rooted at v, each identified by choosing $d-1$ generators from S. Namely:

$$\big\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d'-1\big\}$$

For each check, the bits that connect to it, are given by restricting the coboundary of the check to bits domain of Stab_v:

$$\big(\partial\theta_{\mathbf{v},\mathbf{S'}}\big)|_{\mathrm{Stab}_v} = \big(\sum_{g\in S/S'} \theta_{\mathbf{v},\mathbf{S'\cup g}} + \sum_{g\in S'} \theta_{\mathbf{g^{-1}v},\mathbf{S'\cup g}}\big)|_{\mathrm{Stab}_v} = \sum_{g\in S/S'} \theta_{\mathbf{v},\mathbf{S'\cup g}}$$

2. Checks are rooted at the positive sibling of v. Consider such a sibling, let's say gv (namely the vertex that is given by stepping in the direction $g \in S$ from v).

Notice that any face rooted at gv that contains ggv cannot be contained in faces rooted at v. Thus, the checks of gv correspond to choosing subsets from S/g. Namely:

$$\bigcup_{g\in S}\left\{\theta_{\mathbf{gv},\mathbf{S'}} : S'\subset S/g, |S'| = d'-1\right\}$$

And each check is supported by:

$$\begin{aligned}\left(\partial\theta_{\mathbf{gv},\mathbf{S'}}\right)|_{\mathrm{Stab}_v} &= \Big(\sum_{g'\in S/S'}\theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}} + \sum_{g'\in S'}\theta_{\mathbf{g'^{-1}gv},\mathbf{S'}\cup\mathbf{g'}}\Big)|_{\mathrm{Stab}_v}\\ &= \Big(\sum_{\substack{g'\in S/S'\\ g'\neq g}}\theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}}\Big) + \theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

3. Checks that their roots are at a two-step distance from v, namely vertices of the form $g, g' \in S$, where $g \neq g'$. By the arguments presented in the second type, the checks are the $d'-1$ faces which do not have the generators on the path leading from v to them, i.e., g, g', in their generator set.

$$\bigcup_{g,g' in S'}\left\{\theta_{\mathbf{g'gv},\mathbf{S'}} : S'\subset S/\{g,g'\}, |S'| = d'-1\right\}$$

And for each check, its support is:

$$\begin{aligned}\left(\partial\theta_{\mathbf{g'gv},\mathbf{S'}}\right)|_{\mathrm{Stab}_v} &= \Big(\sum_{g''\in S/S'}\theta_{\mathbf{gg'v},\mathbf{S'}\cup\mathbf{g''}} + \sum_{g\in S'}\theta_{\mathbf{g''^{-1}gg'v},\mathbf{S'}\cup\mathbf{g''}}\Big)|_{\mathrm{Stab}_v}\\ &= \theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}} + \theta_{\mathbf{g'v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

Definition 6.4 (Small Code at Vertex v)**.** For a vertex v, we denote by $\mathcal{C}_v$ the *small code at vertex* v, which is defined to be the binary assignments over the bits (d' faces) rooted at v that satisfy the positive checks of v, namely:

$$\begin{aligned}\mathcal{C}_v := \big\{z : \partial^- z = 0 \text{ and }\\ z\in \mathrm{Span}\big(\{\theta_{\mathbf{v},\mathbf{S'}} : S'\subset S, |S'| = d'\}\big)\big\}\end{aligned}$$

The checks of the code are exactly the first type checks of Stab_v.

Claim 6.5. Assume $d = 4$ and $d' = 2$. Let v be a vertex. For any $z \in \mathcal{C}_v$, there is a stabilizer $w \in \mathrm{Stab}_v$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v.

Proof. We argue that the top six rows of the generating matrix of Stab_v are exactly the generating matrix of $\mathcal{C}_v$. This means that any codeword of Stab_v is an extension of a codeword of $\mathcal{C}_v$. Denote by $H_{\mathcal{C}_v}$ and H_{Stab_v} the parity check matrices of $\mathcal{C}_v$ and

Stab_v respectively. Similarly, denote by $G_{\mathcal{C}_v}$ and G_{Stab_v} their generating matrices. The parity check matrices are:

$$H_{\mathcal{C}_v} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \qquad H_{\mathrm{Stab}_v} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Where we mark in blue, brown, and red the first, second, and third types of checks.

$$G_{\mathcal{C}_v} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \qquad G_{\mathrm{Stab}_v} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

□

Conjecture 6.6. For any integers d and $d' < d$, $G_{\mathcal{C}_v}$ is contained in G_{Stab_v}. In particular, Claim 6.5 holds for any dimension.

Claim 6.7. Let z be a finite assignment. Suppose that

$$\max\left\{L_{d+1}(u) : u \in z\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1$$

Proof. Let $v_1, v_2, \ldots, v_k$ be the vertices in z that achieve the maximum for L_{d+1}, namely $v_1, v_2, \ldots, v_k = \arg\max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial^{\pm} z$ for any $i \in [k]$.

Consider the ith vertex v_i. By definition, since $v_i \notin \partial^{\pm} z$, we have $\partial^{\pm} z|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z, it is also a local maximum, and therefore $z|_{v_i+} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v_i, namely $z|_{v_i} \in \mathcal{C}^{\pm}_{v_i}$. Thus, by Claim 6.5, there is a stabilizer $w_i \in \mathrm{Stab}^{\pm}_{v_i}$ such that w_i and z agree on the positive faces of v_i, namely $w_i|_{v_i} = z|_{v_i}$.

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{C_X} z$. In addition, for any $j \neq i$, it follows that $v_j \notin w_i$. Otherwise, there exists $S' \subset S$ such that $v_j \in \theta_{v_i, S'}$, and therefore $L_{d+1}(v_j) > L_{d+1}(v_i)$, in contradiction to v_i and v_j being both maximum points for L_{d+1} in $\partial^{\pm} z$. Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{z}$. Hence, the vertex domain of $\tilde{z}$ is contained in the union of the vertices of z with the vertices of $w_1, w_2, \ldots, w_k$, substituting $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} &\leq \max\left\{L_{d+1}(u) : u \in z \bigcup_i w_i / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1\end{aligned}$$

□

Claim 6.8. Let z be a finite assignment. Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Proof. Let z be a finite assignment, and let k be an integer such that $k \geq 1$ and

$$\max\left\{L_{d+1}(u) : u \in z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} + k$$

For an integer $l \leq k$, consider the assignments $\tilde{z}_0, \tilde{z}_1, .., \tilde{z}_l$ defined as follow:

- The first element, $\tilde{z}_0 = z$.
- If the $i-1$th element satisfies

  $$(\star) \quad \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} \tilde{z}_{i-1}\right\}$$

  Then define the ith element, $\tilde{z}_i$, to be the result of applying the construction, given in Claim 6.7, on $\tilde{z}_{i-1}$, namely $\partial \tilde{z}_{i+1} = \partial z_i$ and

  $$\max\left\{L_{d+1}(u) : u \in \tilde{z}_i\right\} \leq \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} - 1$$

  If the $i-1$th element doesn't satisfies $\star$, then set $l = i - 1$.
- If the kth element was defined, set $l = k$.

Since $\sim_{C_X}$ is a transitive relation, we have that for any $i \in [l]$, $z \sim_{C_X} \tilde{z}_i$. If $l \neq k$, then there exists $\tilde{z}_i$ such that $\tilde{z}_i \sim_{C_X}$ and $\tilde{z}_i$ satisfies $\star$, namely by setting $\tilde{z} = \tilde{z}_i$, we get the desired. So, it is left to handle the case where $l = k$. In that case, set $\tilde{z} = \tilde{z}_k$. By Claim 6.7:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Yet, since $\tilde{z} \sim_{C_X} z$, it follows that $\partial \tilde{z} = \partial z$, So:

$$\max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\}$$

Therefore, we have the equality:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

□

Claim 6.9. Let z be a finite assignment and let ξ be the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$: $\max\left\{L_i(v) : v \in \partial z\right\} = \max\left\{L_i(v) : v \in \partial \xi\right\}$
2. And for $i = d+1$: $\max\left\{L_{d+1}(v) : v \in \partial z\right\} > \max\left\{L_{d+1}(v) : v \in \partial \xi\right\} + 1$

Proof. First, since the Sweep rule is invariant to addition by stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z. Denote by ϕ the correction, by $\tilde{\xi}$ the result of applying the Sweep rule on $\tilde{z}$, and by w the stabilizer which is the difference between z and $\tilde{z}$, namely:

$$\begin{aligned}\xi &= z + \phi \\ \tilde{\xi} &= \tilde{z} + \phi \\ \tilde{z} &= z + w\end{aligned}$$

The $\partial^{\pm}$ boundary of $\tilde{\xi}$ equals:

$$\partial\tilde{\xi} = \partial\left(\tilde{z} + \phi\right) = \partial\left(z + \phi\right) = \partial\xi$$

So, in total, we have that for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, it holds that $\partial\tilde{z} = \partial z$ and $\partial\tilde{\xi} = \partial\xi$. Therefore, showing that there exists $\tilde{z} \sim_{C_X} z$ for which the claim holds proves the claim for z. By Claim 6.8, there is $\tilde{z} \sim_{C_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\}$$

For such $\tilde{z}$, any maximum of L_{d+1} in $\partial\tilde{z}$, is also a maximum in $\tilde{z}$. Denote by $v_1, .., v_k$ the maximum points in $\partial\tilde{z}$. Since they are maximum points in $\tilde{z}$, it holds that for each $i \in [k]$, $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$, and hence $\partial\tilde{z}|_{v_i} = \partial\tilde{z}|_{v_i+}$. Since $\tilde{z}|_{v_i+}$ is an assignment rooted at v_i, then v_i can suggest $\tilde{z}|_{v_i+}$, namely the condtion in Sweep rule is satisfied for vertex v_i. Thus:

$$\partial\tilde{\xi}|_{v_i} = \partial\tilde{z}|_{v_i} + \partial\phi|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{\xi}$. For any vertex v denote by ϕ_v the correction made by the vertex v, namely $\phi = \sum_{v\in\partial\tilde{z}} \phi_v$. Hence, the vertex domain of $\tilde{\xi}$ is contained in the union of the vertices of $\tilde{z}$ with the vertices of $\cup_{v\in V}\phi_v$, substituting maximum points $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \partial\tilde{\xi}\right\} &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z} \bigcup_{v\in\partial\tilde{z}} \phi_v / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z}\right\} - 1\end{aligned}$$

Let v be a vertex that is a maximum of L_g in $\partial\tilde{z}$, maximum points of L_g do not see syndrome at the g direction. If and:

$$\phi_v = \sum \theta_{\mathbf{v},\mathbf{S_i'}}$$

By Claim 6.2 $\partial\tilde{z}|_{gv} = 0$, Hence:

$$\begin{aligned}\partial\phi_v|_{gv} &= \left(\partial \sum \theta_{\mathbf{v},\mathbf{S_i'}}\right)|_{gv} \\ &= \left(\sum_{S_i'}\sum_{g'\in S_i'} \theta_{\mathbf{v},\mathbf{S_i'/g'}} + \theta_{\mathbf{g'v},\mathbf{S_i'/g'}}\right)|_{gv} \\ &= 0\end{aligned}$$

Thus if $g \in \cup S_i'$, Then:

$$\theta_{\mathbf{gv},\mathbf{S_i'/g}} \in \partial\phi_v|_{gv}$$

So for $\theta_{\mathbf{gv},\mathbf{S_i'/g}} = 0$, it must appear at least twice in the sum, yet the left terms in the sum cannot contain it (they differ by the root vertex), and if there is a right term $\theta_{\mathbf{gv},\mathbf{S_j'/g}} = \theta_{\mathbf{gv},\mathbf{S_i'/g}}$, then it implies that $S_j = S_i$. Which is a contradiction, therefore: $g \notin \cup S_i'$ and $gv \notin \phi_v$.

$$\begin{aligned}\max\left\{L_g(u) : u \in \partial\tilde{\xi}\right\} &\leq \max\left\{L_g(u) : u \in \partial\tilde{z} \bigcup_{v\in\partial\tilde{z}} \phi_v / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_g(u) : u \in \partial\tilde{z}\right\} + 1 - 1\end{aligned}$$

□

6.1 The Sweep-rule Phase In The Dual Code

Claim 6.10. Let z be an assignment. Then:

1. For any generator $g \in S$, and potential wall L_g it holds that:

$$\max_{v'\in\partial^- z} L_g(v') \geq \max_{v'\in\partial^+\phi(z)} -L_g(v')$$

2. The following equality holds:

$$\max_{v'\in\partial^- z} L_{d+1}(v') = \left(\max_{v'\in\partial^+\phi(z)} -L_{d+1}(v')\right) + d' - 1$$

Proof. For the first part, let u be a local maximum of $-L_g$ in $\partial^+\phi(z)$. Let v be a vertex and let $S' \subset S$ be a subset of generators, of size $d'-1$, such that $\theta_{\mathbf{v},\mathbf{S'}}$ is an unsatisfied check, induced by $\phi(z)$, containing u. Let's split into cases, depending on whether $g \in S'$.

1. Assume that $g \notin S'$; thus, $L_g(v) = L_g(u)$. Since ϕ-transpose sends checks to checks, we have that the check $\partial^-\theta_{\mathbf{v^{-1}},\mathbf{S/S'}} = \phi(\partial^+\theta_{\mathbf{v},\mathbf{S'}})$ is not satisfied, and therefore any vertex $w \in \theta_{\mathbf{v^{-1}},\mathbf{S/S'}}$ also belongs to $\partial^-(z)$. Combining that $g \notin S'$ and therefore $g \in S/S'$, we get that $gv^{-1} \in \partial^-(z)$. In particular:

$$\max_{v'\in\partial^- z} L_g(v') \geq L_g(gv^{-1}) = 1 + L_g(v^{-1}) = 1 - L_g(u)$$

2. Assume that $g \in S'$. Then $u \neq gv$ (otherwise $-L_g(u) < -L_g(v)$), and again $L_g(u) = L_g(v)$. By repeating the argument above, we get that $v^{-1} \in \partial^- z$, and therefore:

$$\max_{v' \in \partial^- z} L_g(v') \geq L_g(v^{-1}) = -L_g(u)$$

So in overall we get that:

$$\max_{v' \in \partial^- z} L_g(v') \geq \max_{v' \in \partial^+ \phi(z)} -L_g(v')$$

For the second part, let u be a maximum of $-L_{d+1}$ in $\partial^+ \phi(z)$. Let v be a vertex and let $S' \subset S$ be a subset of generators, of size $d' - 1$, such that $\theta_{\mathbf{v},\mathbf{S'}}$ is an unsatisfied check, induced by $\phi(z)$, containing u. Clearly, it must holds that:

$$u = (\prod_{g \in S'} g)v$$

Now $\theta_{\mathbf{v^{-1}},\mathbf{S/S'}}$ is an unsatisfied check in $\partial^- z$, and v^{-1} is a maximum of L_{d+1} in $\partial^- z$. To see this, assume through contradiction that the maximum of L_{d+1} in $\partial^+ z$ is different from v^{-1} and denote it by v'. Then there is an unsatisfied check containing v' where v' is its root. By moving again to the transposed image by ϕ, we find an unsatisfied check induced by $\phi(z)$ which is rooted in v', and the extremum vertex on it has a higher $-L_{d+1}$ than u.

Therefore we get the strict equality:

$$\begin{aligned} \max_{v' \in \partial^- z} L_{d+1}(v') &= L_{d+1}(v^{-1}) = -L_{d+1}(v) = -(L_{d+1}(u) - (d' - 1)) \\ &= \big(\max_{v' \in \partial^+ \phi(z)} -L_{d+1}(v') \big) + d' - 1 \end{aligned}$$

□

Corollary 6.11. Application of the Sweep rule in the dual code changes the extremum values of $\{L_g\}_{g \in S}$ and L_{d+1} exactly as it changes them in the original code. Namely, the extremum values of $\{L_g\}_{g \in S}$ don't increase, and the maximum value of L_{d+1} decreases by one.

6.2 The bottom line

Claim 6.12. The bottom line. **[COMMENT]** Complete it.

7 Toric Encoded Circuits and Toom's Contours

Definition 7.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is a *Toric encoded circuit* if the following holds:

1. Any layer of the circuit consists of classical and quantum registers, such that there is a Toric code $\mathcal{C}$ such that the quantum registers are encoded by the *nice encoding of* $\mathcal{C}$. At time $t = 0$, the quantum registers are initialized to be either $|\bar{0}\rangle$, encoded $|\mathbf{H}\rangle$, encoded $|\mathbf{S}\rangle$, or encoded $|\mathbf{T}\rangle$. The classical registers are initialized to be zeros.

2. The layers of C on the quantum registers at even times are sweep-cycles, and the layers at odd times are realizations of either logical Paulis, **CX**, or measurements in the computational and parity bases. On the classical registers, including the measured register, we allow any two-bit gates.

Definition 7.2 (Base graph G_o)**.** Let C be a Toric encoded circuit at depth d. Denote its quantum registers by $\{R_i^j\}$, where the subscript and the superscript identify the register number and the time, respectively. For example, R_0^0 is the first register at time zero. Notice that measurements take quantum registers into classical, Thus any quantum register has a maximal depth, which we denote by d_i. Let's denote by $\{\mathcal{G}_i\}$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$, as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the disjoin union over the vertices in $\{\mathcal{G}_i\}$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigcup_i \big(V(\mathcal{G}_i) \times [d_i]\big)$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \big\{\{(v,t),(u,t')\} :\ & \text{if} \\ & \text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t+1 \\ & \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1\big\} \end{aligned}$$

[**COMMENT**] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 7.3 (Tooms Contour Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$. We define the *Tooms contour graph* $\mathcal{H} = (V_\mathcal{H}, E_\mathcal{H})$ as follows:

1. The vertices $V_\mathcal{H}$ are the boundary of $\mathcal{D}$. Namely:

$$V_\mathcal{H} := \big\{v_t : v_t = (v,t) \in V_o \text{ and there exists a face } f \in \partial\mathcal{D}_t \text{ such } v \in f\big\}$$

We extend the action of potential walls to act on the vertices of $V_{\mathcal{H}}$ by acting on the associated points in $\mathbb{R}^d$. More concretely, for a vertex $v \in V_{\mathcal{H}}$, whose corresponding vertex on the infinite $\mathbb{R}^d$ Toric is $\tilde{v} \in \mathcal{G}$, and a potential wall L_i, the value of L_i at v is:

$$L_i(v) := L_i(\tilde{v})$$

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where for any logical gate g which applied in time t in $C[\mathcal{E}]$ we add edge to $A_{\mathcal{H}}$ as follow:

   (a) If g is either a logical Pauli, a measurement (both in the computational and phase basis), or an identity, then for any face f in the register on which g acts, we add:

   $$\big\{\{(u,t),(v,t+1)\} : u,v \in f \text{ and } v,u \in V_{\mathcal{H}}\big\}$$

   (b) If g is a logical **CX** then for any physical **CX** in its realization, between faces f_1 into f_2 at time t we add:

   $$\big\{\{(u,t),(v,t+1)\} : u \in f_1 \cup f_2, v \in f_1 \cup f_2 \text{ and } v,u \in V_{\mathcal{H}}\big\}$$

   In addition for any fault g of $\mathcal{E}$, that is applied at time t, and any face $f \in \mathrm{Locc}(g)$, we add the edges:

   $$\big\{\{(u,t),(v,t+1)\} : u,v \in f \text{ and } u,v \in V_{\mathcal{H},X}\big\}$$

   To construct $F_{\mathcal{H}}$, we connect any pair of vertices (v,t), (u,t) if there exists a vertex $(w,t+1)$ such both of them are connected to it trough $A_{\mathcal{H}}$. Namely:

   $$F_{\mathcal{H}} = \big\{e = \{u,v\} \in E_o : \text{ exists } w \in V_{\mathcal{H}} \text{ such } \{v,w\},\{u,w\} \in A_{\mathcal{H}}\big\}$$

We call $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$ the *arrows* and *forks* edges respectively.

For the analyses we would like to have the following definition:

Definition 7.4. For a Toric encoded quantum circuit C, a set of faults $\mathcal{E}$, and a step τ, according to the standard order, let $C[\mathcal{E}]_\tau$ be the partial computation circuit of $C[\mathcal{E}]$ up to step τ. We define the *analysis Toom's contour at step* τ $\mathcal{H}_\tau = (V_{\mathcal{H}}, E_{\mathcal{H}})$ to be the graph obtained by taking the Toom's contour of $C[\mathcal{E}]_\tau$ and adding vertices and edges as follows:

1. For vertices, we define $t^\star$ as the maximum time of vertices up to step τ plus one, and then add the subset:

   $$\big\{v_{t^\star} : v_{t^\star} = (v,t^\star) \text{ and there exists a } d' \pm 1 \text{ dimensional face } f \in \partial D_{t^\star-1}$$
   $$\text{and a vertex } u_{t^\star-1} \text{ such } v_{t^\star}, u_{t^\star-1} \in f\big\}$$

2. For the edges, we add the following arrows:

$$\Big\{\big\{(u,t^\star-1),(v,t^\star)\big\} \text{ if there exists a } d'\pm 1 \text{ dimensional face } f\in\partial D_{t^\star-1} \text{ and a vertex such } v_{t^\star},u_{t^\star-1}\in f\Big\}$$

3. Finally, we add forks for vertices at time $t^\star$ to preserve the property that any two vertices with the same time coordinate, which are connected to a third via arrows, are also connected by a fork.

7.1 Space-Time Graph

Definition 7.5. Consider a graph $G=(V,A,F)$, where V is a set of vertices, and A and F are sets of undirected edges. For $v\in V$, denote by $\mathrm{pre}(v)=\{u\in V\colon \{u,v\}\in A\}$, we extend the notion for subsets of vertices $X\subset V$ by $\mathrm{pre}(X)=\bigcup_{v\in X}\mathrm{pre}(v)$. Let $T\colon V\to\mathbb{N}$, We say that (G,T) is a time graph if and only if:

1. $\{x,y\}\in A\Rightarrow T(y)=T(x)+1$
2. For $x,y\in V$ $\{x,y\}\in F$ if and only if there exists $z\in V$ such both $\{x,z\},\{y,z\}\in A$.

We will call to T time-function, to A arrows, and to F forks. Observes that from the second requirement it follows that forks can connect only vertices that have the same time value (set by T).

Remark 7.6. The Toom's contour graph defined in Definition 7.3 is, by definition of it's construction, a time-graph.

Definition 7.7 (History, slice.)**.** For a time graph (G,T) and a vertex $u\in V$, let G_u be the subgraph of (V,A) induced by $\{v\in V\colon T(v)\le T(u)\}$. We will denote by H_u the 'History of u', which is the set of vertices that are connected to u in G_u. A slice is an equivalence class of the relation $u\equiv v$ if and only if $H_u=H_v$. Notice that the time function has to be constant on a slice, to see this consider vertices x,y with $T(x)>T(y)$, thus $x\notin G_y$ and therefore $x\in H_x/H_y$ So they are belong to different classes. Thus, we can extend the time function and the history to slices and write for a slice K, $T(K)$ and H_K.

Claim 7.8. Consider slices K_1,K_2 such that $T(K_1)=T(K_2)$, then either $K_1=K_2$ or H_{K_1},H_{K_2} are disjoint.

Proof. Assume a non-empty intersection, so there exists $z\in H_{K_1}\cap H_{K_2}$ such that for any $x\in K_1$ and $y\in K_2$, there exist arrow-paths $z\to x$ and $z\to y$. Thus, for any $w\in H_{K_1}$, one can concatenate a path $w\to x\to z\to y$ and conclude that $y\in H_{K_2}$. □

Definition 7.9 (The graph of slice)**.** Let K be a slice. The graph of K, denoted by $G_K=(V_K,E_K)$, is the graph whose vertices are slices $R\subset H_K$ with $T(R)=T(K)-1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S,R\in V_K$ are connected in G_K if and only if there is $\{s,r\}=f\in F$ such that $s\in S$ and $r\in R$.

Claim 7.10. G_K is connected.

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider such a path that is minimal in the number of points x belong to it, with $T(x) = T(K)$. By induction on the number k of such points we prove that R and S are connected in G_K.

1. Base. $k = 0$, In this case, the path from r to s lies within $G_r (= G_s)$. Hence $H_r = H_s$, So $R = S$.

2. Induction step. There exists y on the path from r to s such that $T(y) = T(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:

   (a) $\{x, z\} \subset \text{pre}(y) \Rightarrow \{x, z\} \in F$.

   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

   Since the path connecting r, x in H_K has $k - 1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

□

Definition 7.11 (Spanned Set)**.** Consider a subset $A \subset \mathbb{R}^d$, and $d + 1$ points $x_1, \ldots, x_{d+1}$. We say that $\langle A, x_1, \ldots, x_{d+1} \rangle$ is a spanned set of A if for any $i \in [d + 1]$ and $s \in A$, it holds that $L(s) \leq L_i(x_i)$ and in addition there are $v_1, ..v_{d+1} \in V$, not necessary different, that achieve the equalities $L_i(x_i) = L_i(x_i)$. We name $x_1, \ldots, x_{d+1}$ the poles of A. Furthermore, we extend the notation to a spanned slice when A is a slice of some time-graph. Notice that saying $\langle A, x_1, \ldots, x_{d+1} \rangle$ is a spanned set means that $\textbf{Size}(A) = \textbf{Size}(\{x_1, \ldots, x_{d+1}\})$.

Observe that in the case where the subset A is a slice. Each of these poles can be associated with a vertex of G that gives the equality. If there are multiple vertices whose give equality, associate the point with one of them arbitrarily. We will use the notion of applying functions originally defined over vertices to the poles, referring to the application over the vertices associated with them. For example $\text{pre}(x_i) = \text{pre}(v)$ for the vertex $v \in A$ which satisfies $L_i(v) = L_i(x_i)$ and therefore is associated with the pole x_i. Another example is referring to x_i as a vertex, so when saying 'For $x_i \in V_k$' or 'For $\{x_i, u\} \in F$', we refer to the associated vertex or the edge connecting it to u.

Claim 7.12. Consider a time graph (G, T), Let K be a slice in G, and let $\hat{K} = \langle K, x_1, ..x_{d+1} \rangle$ be a spanned slice of K such that $K \neq H_K$. Then for some integer k there exist spanned slices $\hat{K}_0 = \langle K_0, \cdot \rangle, .., \hat{K}_k = \langle K_k, \cdot \rangle$ and Forks $F_1, .., F_k$ such that:

1. For $i \in [d+1]$ $\text{pre}_i(x_i)$ belongs to the poles $\hat{K}_0, .., \hat{K}_k$.
2. Introduce an edge between $\hat{K}_i$ and $\hat{K}_j$ if and only if one of the Forks $F_1, .., F_k$ consists of a pole of $\hat{K}_i$ and a pole of $\hat{K}_j$. Then the graph formed from $\hat{K}_0, .., \hat{K}_k$ is connected.
3. $\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right)$

Proof. First Notice that K doesn't contain leaves in (V, A), since a leaf is connected only to itself via arrows, and therefore its 'slice' equals to its 'History'. Thus $\text{pre}_i(x_i)$ is not empty for $i = 1, .., d+1$. Next consider the graph $G_K = (V_K, E_K)$ from definition 7.9. Since $\{x_i, \text{pre}_i(x_i)\}$ is an arrow $\text{pre}_i(x_i)$ belongs to a slice in V_K. for $i = 1, .., d+1$ denote by R_i the slice in V_K that contains $\text{pre}_i(x_i)$.

By claim 7.10 G_K is connected and therefore there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\text{pre}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $\hat{K}_0, .., \hat{K}_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $\hat{F}_1, \cdots, \hat{F}_k$ in such way that:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(\hat{F}_i) \geq \sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right)$$

This will assure (3).

Let us denote by $J_0, .., J_{2k}$ the enumerated union $K_0, ..K_k, F_1, .., F_k$. Observes that for any choice of $(d+1)(2k+1)$ elements $z_0^1, z_0^2, z_0^3, .., z_0^{d+1} .., z_{2k}^{d+1}$ in $\mathbb{R}^d$, such that $z_j^i \in J_j$ it holds that:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(\hat{F}_i) \geq \sum_{j,i} L_i(z_j^i) \tag{1}$$

Later, we are going to take the points $\{z_j^i\}$ from the intersection of J_j's. For convenience, let us assume that for the index subset $X \subset [2k] \cup \{0\}$, the intersection $\bigcap_{j \in X} J_j$ returns a single point. If there is more than one, then decide on the point in a way that all the chosen points simultaneously satisfy that for any index subsets Y, X, such that $X \cap Y$ and they both introduce a non-trivial intersection $\bigcap_{j \in Z} J_j : Z \in \{X, Y\}$, then:

$$\bigcap_{j \in X} J_j = \bigcap_{j \in Y} J_j$$

Now consider a non-trivial maximal intersection $\bigcap_{j \in X} J_j$, the intersection is maximal in the sense that for any $j' \notin X$ it holds that $.\bigcap_{j \in X \cup \{j'\}} J_j = \emptyset$. We claim that for any i and for any such X, there is $t \in X$ such that J_t is the successor of J_j on

the path to R_i for any $j \in X/\{t\}$. Assume not. Let $x, y \in X$, and denote the paths from J_x, J_y to the ith pole by $\langle J_x, J_{x_2}, \ldots, R_i\rangle$, $\langle J_y, J_{y_2}, \ldots, R_i\rangle$, such that $J_y \neq J_{x_2}$. If $y = x_2$, $y_2 = x$, or $x_2 = y_2 \in X$, then pick other x, y (if there is no such, we get an immediate contradiction). In that case, $\langle J_x, J_y, J_{y_2}, \ldots, R_i\rangle$ is a path from J_x to R_i which is different from $\langle J_x, J_{x_2}, \ldots, R_i\rangle$. Therefore, there are at least two paths from J_x to the ith pole, namely, there is a cycle in contradiction to the minimality of $K_0, \ldots, K_k$.

We will choose the z_j^i by the following process. Pick a z_j^i that has not been set yet. If $J_j = R_i$, then set $z_j^i \leftarrow \mathrm{pre}_i(x_i)$ - we call this a type-1 assignment. Otherwise, consider the path from J_j to R_i (by connectivity, it has to be such a one, and by minimality, there is exactly one). Let J_t be the successor on the path, then pick $z_j^i \in J_j \cap J_t$ - similarly, we call this assignment a type-2 assignment. Since any type-2 assignment is associated with a non-trivial intersection, and any non-trivial intersection induces a type-2 assignment for all i's (since for all i's, one of the J_j on its support is a successor of the rest on their path to the ith pole), we have that:

$$\begin{aligned}\sum_{j,i} L_i(z_j^i) &= \sum_{j,i,z_j^i\in\text{type-1}} L_i(z_j^i) + \sum_{j,i,z_j^i\in\text{type-2}} L_i(z_j^i) \\ &= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right) + \sum_{X\colon \bigcap_{j\in X} J_j\neq\emptyset} (|X|-1)\sum_{i}^{d+1} L_i\left(\bigcap_{j\in X} J_j\right) \\ &= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right)\end{aligned}$$

Plugging it into eq. (1) gives the desired. □

Claim 7.13. The maximal size of a fork is $2d' = d$.

Proof. Consider the rules of forks connecting in Definition 7.3. For forks obtained by rules (a), (b), or as a result of a fault, it's clear that their vertices at their ends are on intersecting faces (when projecting the vertices on $\mathcal{G}$). The maximal fork, in those cases, has the form of:

$$F = \{v, \left(\prod_{g\in S_1'} g\right)\left(\prod_{g\in S_2'} g\right)v\}$$

Where S_1' and S_2' are subsets of S of size d', and therefore the size of such forks is $\mathbf{Size}(F) = 2d'$. □

Definition 7.14. Let $\hat{K}$ be a spanned slice, and let $\hat{K}_0, \ldots, \hat{K}_k$ be spanned slices in H_K, and let $F_1, \ldots, F_k$ be spanned forks, such that $\hat{K}_0, \ldots, \hat{K}_k, F_1, \ldots, F_k$ satisfy the conditions in Claim 7.12. We call $\hat{K}_0, \ldots, \hat{K}_k, F_1, \ldots, F_k$ the *predecessor decomposition* of $\hat{K}$.

Furthermore, we say that $\hat{K}$ is a result of a *sweep-cycle*, *faults*, or a *logical operation* if the layer at time $T(\hat{K}) - 1$ consists of sweep-cycle gates, fault gates (whose source is $\mathcal{E}$), or either measurement or logical Pauli. The *order* of the slice is its time modulo 4,

namely $\text{order}(\hat{K}) = T(\hat{K}) \mod 4$. Observe that slices that are results of faults have an order that is either 0 or 2, whereas slices that are results of a sweep-cycle or logical operation are of orders 1 and 3, respectively.

Corollary 7.15. Consider a spanned slice $\hat{K}$, at finite size, such that $K \neq H_K$. Since $K \neq H_K$, it follows that $\hat{K}$ has a predecessor decomposition, denoted by $\hat{K}_0, \ldots, \hat{K}_k, F_1, \ldots, F_k$. The inequality in Claim 7.12 becomes:

1. If $\hat{K}$ is a result of ξ-sweep-cycle, then:
$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K}) + \xi$$
2. If $\hat{K}$ is a result of either faults or a logical operation, then:
$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K})$$

Proof. For the sweep-cycle part, we use Claim 6.12, which states that after each application of the sweep-cycle over a finite size assignment, the maximal L_{d+1} value of the resulting boundary decreases by 1, while the values of each L_g do not increase.

When $\hat{K}$ is a result of either faults[2] or a logical operator, we use the fact that since the operators are transversal, we have that the support of $\hat{K}$ is contained in the support of $\cup_0^k \hat{K}_i$, and therefore for any $i \in [d+1]$:
$$\max\{L_i(v) : v \in \hat{K}\} \leq \max\{L_i(v) : v \in \bigcup_{i=0}^{k} \hat{K}\}$$

□

Claim 7.16. For a step τ, let $\mathcal{H}_\tau$ be analysis Toom's contour at step τ, and let $t^\star$ be the maximal time at that contour. For a spanned slice $\hat{K}$, at time $T(K) = t^\star$, where $K \neq H_K$, the inequality in Claim 7.12 becomes:
$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(K) - d - d' - 1$$

Proof. We allow vertices at the final time layer to be connected by an arrow if they are at a distance of $d' + 1$ from each other. Therefore, for any $i \in [d]$, the value of L_i for v_i might be greater by 1 than the value of $\text{pre}_i(v_i)$, and the L_{d+1} for v_{d+1} might be greater by $d' + 1$ than the value of the $d + 1$ pole of $\hat{K}$. Hence, overall, we get that:
$$\begin{aligned}\sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right) &\geq \sum_{i=1}^{d} \big(L_i\left(v_i\right) - 1\big) + L_{d+1}(v_{d+1}) - \max d' - 1 \\ &= \mathbf{Size}(K) - d - d' - 1\end{aligned}$$

□

[2]Notice that $\hat{K}$ does not consist of faults since otherwise $K = H_K$, so the meaning here is that $\hat{K}$ contains vertices which weren't flipped in the fault layer.

8 Explantation of a Spanned Slice

Definition 8.1. Consider a Tooms contour graph induced by Toric encoded circuit, with sweep-cycle parameter $\xi = 1$, and let α, β be constants and $\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3 \in \mathbb{A}$ be quadruple, defining such:

$$\alpha := 40(d+1)^2, \quad \beta := 5(d+1)$$
$$\mathcal{X}_r := 1 - r\frac{d+1}{\alpha} = 1 - \frac{r}{40(d+1)}$$

Given sets W of points, A of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $A \cup F$. The sets W, A and F form an explanation of a spanned slice $\hat{K} = \langle K, v_1, .., v_{d+1} \rangle$ of order $\text{order}(\hat{K}) = r$, if and only if:

1. $W \cup \{v_1, .., v_{d+1}\} \subset U \subset H_K$ and the graph $(U, A \cup F)$ is connected.
2. All elements of W are leaves.
3. For $m = |W|$ and $s = \mathbf{Size}(K)$ we have $|A| \leq \alpha(m - \mathcal{X}_r) - \beta s$ and $|F| < m - 1$.

Furthermore, for a step τ and a spanned slice $\hat{K}$ in the analysis Toom's contour at step τ, we extend the definition of explanation as above, but require the following weaker inequality instead: For $m = |W|$ and $s = \mathbf{Size}(K)$, we have $|A| \leq \alpha(m+1) - \beta s$ and $|F| < m - 1$.

Claim 8.2. Every spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ has an explanation.

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. For spans derived from the standard Toom's contours (namely not a analysis Toom's contours), we prove by induction on h.

1. Base: $h = 0$, meaning T is constant on H_K, so no arrows connect points of H_K. Thus, H_K, as well as $\mathbf{K}$, consists only of a single leaf, namely $\hat{K}$ is a result of a fault, and $r = \text{order}(\mathcal{X}_r) \in \{0, 2\}$. In that case, there is a single vertex $u \in V_{\mathcal{H}}$ such that $U = W = \{u\}$, so $m = 1$, $s = 0$, and $|A|, |F| = 0$. It is easy to check that those values satisfy the requirement.

2. Induction Step: $h \neq 0$ implies $K \neq H_K$. Thus, by claim 7.12, we know that for some k there exist spanned slices $\hat{K}_0, .., \hat{K}_k$ and forks $F_1, .., F_k$ that satisfy claim 7.12. In particular, for $r < 3$ it holds that:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K})$$

   Whereas, for $r = 3$:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K}) + \xi = \mathbf{Size}(\hat{K}) + 1$$

Consider $i \in \{0,\ldots,k\}$, and denote $s_i = \mathbf{Size}(\hat{K}_i)$. By the inductive hypothesis and since $T(K_i) = T(K) - 1$, we know that $\hat{K}_i$ has an explanation formed by a set of leaves $\mathbf{W}_i$, a set of arrows $\mathbf{A}_i$, and a set of forks $\mathbf{F}_i$. Let $m_i = |\mathbf{W}_i|$. Thus, the sets $\mathbf{A}_i$ and $\mathbf{F}_i$ have at most $\alpha(m_i - \mathcal{X}_{r-1}) - \beta s_i$ arrows and $m_i - 1$ forks. We define:

(a) $\mathbf{W} = \cup_{i=0}^{k} \mathbf{W}_i$, Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j, thus by Claim 7.8 $\mathbf{W}$ is a union of disjoint set, Hence W contains $m = \sum_{i=0}^{k} m_i$ leaves.

(b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^{k} \mathbf{F}_i$ with at most $k + \sum_{i=0}^{k} m_i - 1 = m - 1$ elements.

(c) $\mathbf{A} = \{\{v_i, \mathrm{pre}_i(v_i)\}\} \cup_{i=0}^{k} \mathbf{A}_i$. For bounding the number of arrows in $\mathbf{A}$ we condition upon whether $r = 3$, For convenient, let us denote by Y the indicator, which indicate if $r = 3$.

$$
\begin{aligned}
&d + 1 + \sum_{i=0}^{k} \alpha(m_i - \mathcal{X}_{r-1}) - \beta s_i \\
\leq &d + 1 + \alpha m - \mathcal{X}_{r-1}\alpha(k+1) - \beta\left(s + Y\xi - k \cdot \mathbf{Size}(\text{fork})\right) \\
= &\alpha\left(m - \mathcal{X}_r\right) - \beta s + \left(\alpha(\mathcal{X}_r - \mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha - \beta Y\xi + \beta k \cdot \mathbf{Size}(\text{fork})\right)
\end{aligned}
$$

So it's left to show that the term in the closure is negative for all integers k:

$$\alpha(\mathcal{X}_r - \mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha - \beta Y\xi + \beta k \cdot \mathbf{Size}(\text{fork})$$

We do that by showing that for the choice of the parameter values, $\alpha, \beta, \mathcal{X}_0, \ldots, \mathcal{X}_3$, the slope is negative, and then the value of the term at k equals zero is negative. For the slope to be negative, we require:

$$-\alpha\mathcal{X}_{r-1} + \beta\mathbf{Size}(\text{fork}) \leq 0 \Rightarrow \mathbf{Size}(\text{fork}) \leq \frac{\alpha}{\beta}\mathcal{X}_{r-1}$$

For negativity at the origin, $k = 0$, we require for $r < 3$:

$$
\begin{aligned}
(\mathcal{X}_r - \mathcal{X}_{r-1})\alpha + d + 1 &\leq 0 \\
\Rightarrow \mathcal{X}_r &\leq \mathcal{X}_{r-1} - \frac{d+1}{\alpha}
\end{aligned}
$$

And for $r = 3$:

$$
\begin{aligned}
(\mathcal{X}_3 - \mathcal{X}_2)\alpha + d + 1 - \beta\xi &\leq 0 \\
\Rightarrow \mathcal{X}_3 &\leq \mathcal{X}_0 - 3\frac{d+1}{\alpha} + \frac{\beta}{\alpha}\xi
\end{aligned}
$$

So in total:

$$\begin{aligned}
\mathcal{X}_1 &\leq \mathcal{X}_0 - \frac{d+1}{\alpha} \\
\mathcal{X}_2 &\leq \mathcal{X}_0 - 2\frac{d+1}{\alpha} \\
\mathcal{X}_3 &\leq \mathcal{X}_0 - 3\frac{d+1}{\alpha} + \frac{\beta}{\alpha}\xi \\
\mathcal{X}_0 &\leq \mathcal{X}_0 - 4\frac{d+1}{\alpha} + \frac{\beta}{\alpha}\xi \\
\textbf{Size}(\text{fork}) &\leq \frac{\alpha}{\beta}\mathcal{X}_r
\end{aligned}$$

The first three inequalities hold by the definitions of the $\mathcal{X}_r$, the fourth is satisfied since $\frac{\beta}{\alpha}\xi = 4\frac{d+1}{\alpha}$, the last is obtained due to Claim 7.13, that states that the maximal size of a fork equals $4d$, and that $\frac{\alpha}{\beta}\mathcal{X}_r$ is greater than $4d$.

Finally, by Claim 7.12, the graphs induced by $\mathbf{A}_i \cup \mathbf{F}_i$ are connected, thus the graph induced by $\mathbf{A} \cup \mathbf{F}$ is also connected. Therefore, $\mathbf{W}, \mathbf{A}, \mathbf{F}$ form an explanation of K.

It's left to prove for spanned slices in the analysis Toom's contours. Notice, that h has to be greater than zero, We prove the inequality by repeating on the induction bases, where all the spanned slices that are back in the history of K are spanned slices in the standard Toom's contour. So the only different is in the bounding of A's size. We use **??** and get:

$$\begin{aligned}
|A| &\leq \alpha\,(m+1) - \beta s \\
&\quad + (-\alpha(1+\mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha + d + d' + 1 + \beta k \cdot \textbf{Size}(\text{fork})) \\
&\leq \alpha\,(m+1) - \beta s \\
&\quad + (-\alpha + d + 1 - \mathcal{X}_{r-1}k\alpha + d + d' + 1 + \beta k \cdot \textbf{Size}(\text{fork}))
\end{aligned}$$

Where we used the fact that $\mathcal{X}_{r-1} \in (0,1)$ for $r \in \{0,1,2,3\}$. Clearly, the bound is satisfied since $\alpha \geq 2d + d' + 1$ and $\mathcal{X}_{r-1}\alpha \geq \beta \cdot \textbf{Size}(\text{fork})$. □

Corollary 8.3. Let $\hat{K}$ be a spanned slice with size s, and explanation with $|A|$ arrows, and $|F|$ forks. Then it is induced by at least the following number of faults:

$$\frac{1}{2^{d'}} \min\{|F| + 1, \alpha^{-1}(|A| - 1) + \beta s\}$$

Proof. We bound the Claim 8.2 by the number of leaves in the explanation, and use the fact that any d'-face has $2^{d'}$ vertices, so each fault contributes at most $2^{d'}$ leaves to the explanation. □

[COMMENT] Check what happened if we set $\xi = \sqrt{d}, \beta = \sqrt{d}$ is that better?

9 Counting Subtrees.

Claim 9.1. Let $G = (V, E)$ be a $\Delta + 1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. The number of subgraphs rooted at r with less than k edges is at most α^k.

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tress. There is an injection between the subtrees of G, rooted at v, with at most k edges, to the subtrees of T_Δ rooted at v, with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v, with at most k edges to the subtrees of T_2, rooted at v, with at most $k \log \Delta$ edges.

[COMMENT] In the second map, we think on replacing each of the vertex with $\Delta - 1$-leaves binary tree, each leaf is associated with a outgoing edge. .

Thus, it's enough to bound the number of of subtrees in T_2 rooted at v, with at most $k \log \Delta$ edges, that number is less than the number of binary tress with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^{k} \frac{\Delta^{2j}}{(j \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta} \sum_{j=1}^{k} \frac{1}{j^{\frac{3}{2}}} \leq \zeta(\frac{3}{2}) \cdot \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

□

Now, from here we get an upper bound on the number of subgraphs that connected l vertices at the top level. We bound it from above by counting combination of rooted subtrees, such the vertices are their root and the overall edges number is summed up to k. Thus:

$$\leq \zeta(\frac{3}{2}) \binom{k-1}{l-1} \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

10 Tree Separator Lemma

For a tree $T = (V, E)$ and a vertex $v \in V$, we call the subtree obtained by taking the descendants of v the subtree rooted at v.

Claim 10.1. Let $T = (V, E)$ be a tree with $m > 3$ leaves and τ edges, then it has a subtree with $m' \in (\frac{1}{3}m, \frac{2}{3}m)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Proof. Order the vertices according to their heights. Denote by $v \in V$ the lowest vertex for which the subtree rooted at v has at least $\frac{1}{3}m$ leaves. Now, denote by T_1 the subtree obtained through the following iterative process over v's children:

1. Initialize T_1 as the subtree rooted at v.
2. For a child u of v:
   (a) Check if erasing from T_1 the subtree rooted at u doesn't decrease the number of leaves below $\frac{1}{3}m$. If so, then we update T_1 to be the result of the erasure.

Clearly, the algorithm is defined such that T_1 has to have at least $\frac{m}{3}$ leaves. Furthermore, since any child of v is a root of a subtree with fewer than $\frac{m}{3}$ leaves (otherwise we get a contradiction to the minimality of v's height), we have that if T_1 would have more than $\frac{2}{3}m$ leaves, then the subtraction of any of its children would yield a tree with at least $\frac{2}{3}m - \frac{1}{3}m = \frac{1}{3}m$ leaves. Namely, v has in T_1 a child u such that the subtraction of the subtree rooted at u from T_1 results in a subtree with more than $\frac{1}{3}m$ leaves. But if indeed there were such a child, then the algorithm would erase it, so there is not.

Now let $T_2 = T/T_1 \cup v$, namly the tree induced by the subtraction of T_1's edges from T. Observes that since the summation of leaves in T_1 and T_2 is the leaves in T, and T_1 has no more than $\frac{2}{3}m$ leaves, and no less than $\frac{1}{3}$ leaves, it holds that the number of leaves in T_2 is also in the range $(\frac{1}{3}m, \frac{2}{3}m)$. We claim that at least one of the subtrees has fewer than $\frac{m'}{m}\tau$ edges.

To see this, denote by m_1, m_2 and by τ_1, τ_2 the leaves and the edges in T_1, T_2 respectively. Now:

$$\frac{\tau}{m} = \frac{\tau_1 + \tau_2}{m_1 + m_2} = \frac{m_1}{m_1 + m_2}\frac{\tau_1}{m_1} + \frac{m_2}{m_1 + m_2}\frac{\tau_2}{m_2}$$

Namely, $\frac{\tau}{m}$ is a weighted average of $\frac{\tau_1}{m_1}$ and $\frac{\tau_2}{m_2}$, and therefore it is greater than their minimum. Peak the subtree with the minimal quotient. □

By repeating recursively on the above construction, we get the following corollary:

Corollary 10.2. Let $T = (V, E)$ be a tree with m leaves and τ edges. Then, for any $\alpha > 0$, it has a subtree with $m' \in \left(\frac{1}{3}\alpha m, \frac{2}{3}\alpha m\right)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Claim 10.3. Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.

Proof. Since the construction of claim 10.1 preserves the ratio of edges to leaves, we have that for a T' obtained using the construction, the number of edges satisfies $\tau' \leq \gamma m'$. So, it's enough to look for the maximal α such that:

$$\gamma m' \leq \gamma \frac{2}{3}\alpha m \leq \frac{\beta}{2}n \Rightarrow \alpha = \frac{3\beta}{4\gamma}\frac{n}{m}$$

Which gives $m' \geq \frac{1}{3}\alpha m = \frac{\beta}{4\gamma}n$.

□

Claim 10.4. Assume that until step τ, there was no deviation cluster of size greater than d. Let $\hat{K}$ be a spanned slice induced by a deviation of a cluster state at step τ, and let ω be an explanation for $\hat{K}$ such that the number of vertices in ω is greater than βn. Then there is a subgraph (subtree) ω' of ω such that ω' has fewer than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves.

Proof. Let ω be an explanation of $\hat{K}$ with more than βn vertices. Since there is no deviation cluster of size greater than d until step τ, we have that every vertex of ω is a spanned slice induced by a boundary of a deviation cluster of size at most d. Thus, ω satisfies the shrink property in **??**, and by Claim 8.2, the ratio of edges to leaves of ω is at most γ. Observe that a spanning tree of ω has a ratio of edges to leaves that is at most ω's ratio, and the number of edges equals ω's vertices minus 1. Therefore, by applying Claim 10.3 on ω's spanning tree, ω has a subtree with less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves. □

Proof. Now, since any two connected vertices are at a space-distance of at most 1, it follows that all the leaves in the subtree are at a space-distance of at most βn from each other. Thus, if $\beta < 1$, no two of them have the same space-coordinate modulo n. And therefore the probability to have such subtree is at most $p^{\frac{\beta}{4\gamma}n}$.

Because any ω with more than βn vertices implies the existence of a subtree with size less than $\frac{\beta}{2}n$, it's enough to bound the probability of having such a subtree. Considering that the trees are invariant to shifting by $n \cdot \mathbf{1}_i$, we have that it's sufficient to show that there are no such trees which are rooted in the finite cube $\mathbb{Z}_n^d \times \mathbb{Z}_t$. Finally, we use claim 9.1 to bound the number of such trees given a fixed root. Overall, we get:

$$\mathbf{Pr}\big[\text{No } \omega\text{s with more than } \beta n \text{ vertices}\big] \leq n^d t \left(\xi^2 p^{\frac{\beta}{4\gamma}}\right)^n$$

□

11 Proof of The Theorem

Definition 11.1 (Simulation and Logical Errors)**.** For a quantum code $\mathcal{C}$ of length n and a decoder for $\mathcal{C}$, denoted by D, we say that a Pauli operator $P \in \mathcal{P}^{\otimes n}$ is a *logical error* with respect to D if DP does not equal the logical identity.

For quantum circuits C and $\tilde{C}$, a noise channel $\mathcal{E}$, and $p \in (0, 1)$, we say that $\tilde{C}$ *simulates* C with probability $1 - p$ if

1. If there is a mapping between the qubits of C and the registers of $\tilde{C}$, along with a decoding procedure such that, at the end of running $\tilde{C}$, decoding the registers results in the same quantum states as those produced by running C.

2. The probability induced by $\mathcal{E}$ that the deviation on one of the registers has a logical error is less than p.

Theorem 11.2. There exists a constant $p_0 \in (0, 1)$ such that for any local stochastic noise channel $\mathcal{E}$ with rate $p < p_0$, and a quantum circuit C with depth d and width w, there exists a toric encoded circuit $\tilde{C}$ that simulates C with probability $1 - \frac{wd}{n}$.

Proof. By Claim 2.15. □

References

[Den+02] Eric Dennis et al. "Topological quantum memory". In: *Journal of Mathematical Physics* 43.9 (Sept. 2002), pp. 4452–4505. DOI: 10.1063/1.1499754. URL: https://doi.org/10.1063%2F1.1499754.